Human Genome and Medical Biotechnology

Module: SC3303

By

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Lecture 04:

Diagnosis of genetic diseases, earlier detection of genetic predispositions to disease

Diagnosis of Genetic Diseases-Genetic diseases are caused by abnormalities in DNA sequence, structure, or chromosome number. Diagnosis aims to identify the specific genetic alteration responsible for disease, enabling accurate classification, prognosis, and management.

Types of Genetic Diseases

- Single-gene (monogenic) disorders-Caused by mutations in a single gene
- Examples: Cystic fibrosis (CFTR), Sickle cell disease (HBB), Huntington's disease (HTT)

- Chromosomal disorders-Caused by numerical or structural chromosome abnormalities
- Examples: Down syndrome (trisomy 21), Turner syndrome (45,X)

- Multifactorial (polygenic) disorders-Result from multiple genes interacting with environmental factors
- Examples: Diabetes mellitus, hypertension, coronary artery disease

Laboratory Genetic Testing		
Method	Purpose	Examples
Karyotyping	Detect large chromosomal abnormalities	Trisomies, translocations
FISH	Targeted chromosomal changes	Microdeletion syndromes
PCR-based tests	Known gene mutations	Sickle cell disease
Sanger sequencing	Single-gene analysis	Monogenic disorders
Next-Generation Sequencing (NGS)	Multiple genes or whole genome	WES, WGS
Microarray (aCGH/SNP)	Copy number variations	Developmental delay

Introduction:

- Tools for genomic diagnosis have evolved rapidly over the past two decades, resulting in remarkably improved diagnostic rates as well as a significant increase in the number of disease genes identification.

- The core technologies for genetic testing laboratories have evolved from

  - Karyotype analysis
  - Chromosomal microarrays (CMA).
  - FISH
  - Sanger sequencing
  - Next Generation Sequencing (NGS)-based tests (targeted panels, exomes, and genomes)

Introduction:

- Aneuploidy loss or gain of entire chromosomes (for example a human cell having 45 or 47 chromosomes instead of the usual 46).

- Copy number variants, CNVs or smaller regions of DNA, to changes in the structure of the genome (i.e., translocations, inversions, insertions).

- Single nucleotide variants changes in the sequence of the genome (i.e., and short insertions/deletions).

Introduction:

- These advances have had a profound impact on how we diagnose patients who present with clinical features of known genetic disorders.

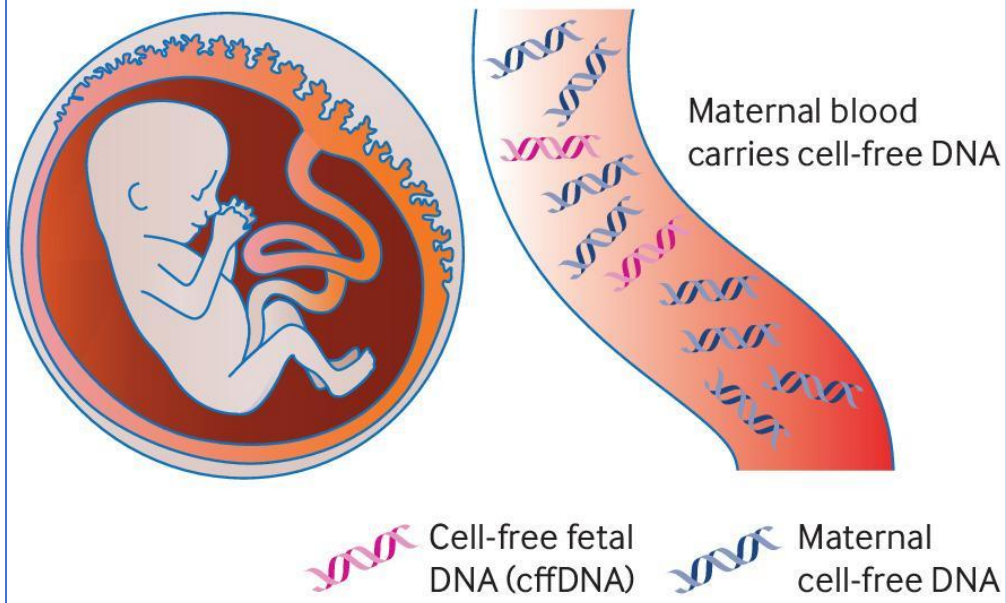
- In the realm of cytogenetics, traditional techniques such as cytogenetic analysis of banded chromosomes are still warranted for some conditions (e.g., trisomies, Turner syndrome).

- However, classic cytogenetic deletion syndromes (such as Wolf-Hirschhorn and Cri-du-Chat syndromes) are now better tested by CMA, with the opportunity to identify smaller deletions or duplications including precise breakpoints.

Noninvasive prenatal testing (NIPT), sometimes called noninvasive prenatal screening (NIPS), is a method of determining the risk that the fetus will be born with certain genetic abnormalities.

- This testing analyzes small fragments of DNA that are circulating in a pregnant woman's blood.

SAMPLES	Fetal DNA + Maternal DNA
TEST	✓ chromosomal abnormalities ✓ CNVs ✓ microdeletion ✓ SNPs
METHODS	entire genome → specific regions → SNPs analysis • massively parallel sequencing • chromosome selective sequencing • multiplex PCR
DATA ANALYSIS	counting, genotyping



Advantages

- Excellent screening test for trisomies 21, 18, and 13
- Low false positive rate gives potential to avoid an invasive test, which carries a risk of miscarriage
- Clarity of reported results: low chance or high chance of aneuploidy

Limitations

- May be a false negative or no result with a low fetal fraction
- May be a false positive with "vanishing" twin, maternal cancer, or placental mosaicism
- Cost of assay

Chromosomal Microarray Analysis (CMA)

- Chromosomal microarray analysis (CMA) is a technology used for the detection of clinically-significant microdeletions or duplications, with a high sensitivity for submicroscopic aberrations.
- Because chromosomal microarray analysis has a greater resolution than conventional karyotyping, it can detect deletions and duplications down to a 50- to 100-kb level.

- There are two major type of arrays to detect CNVs,

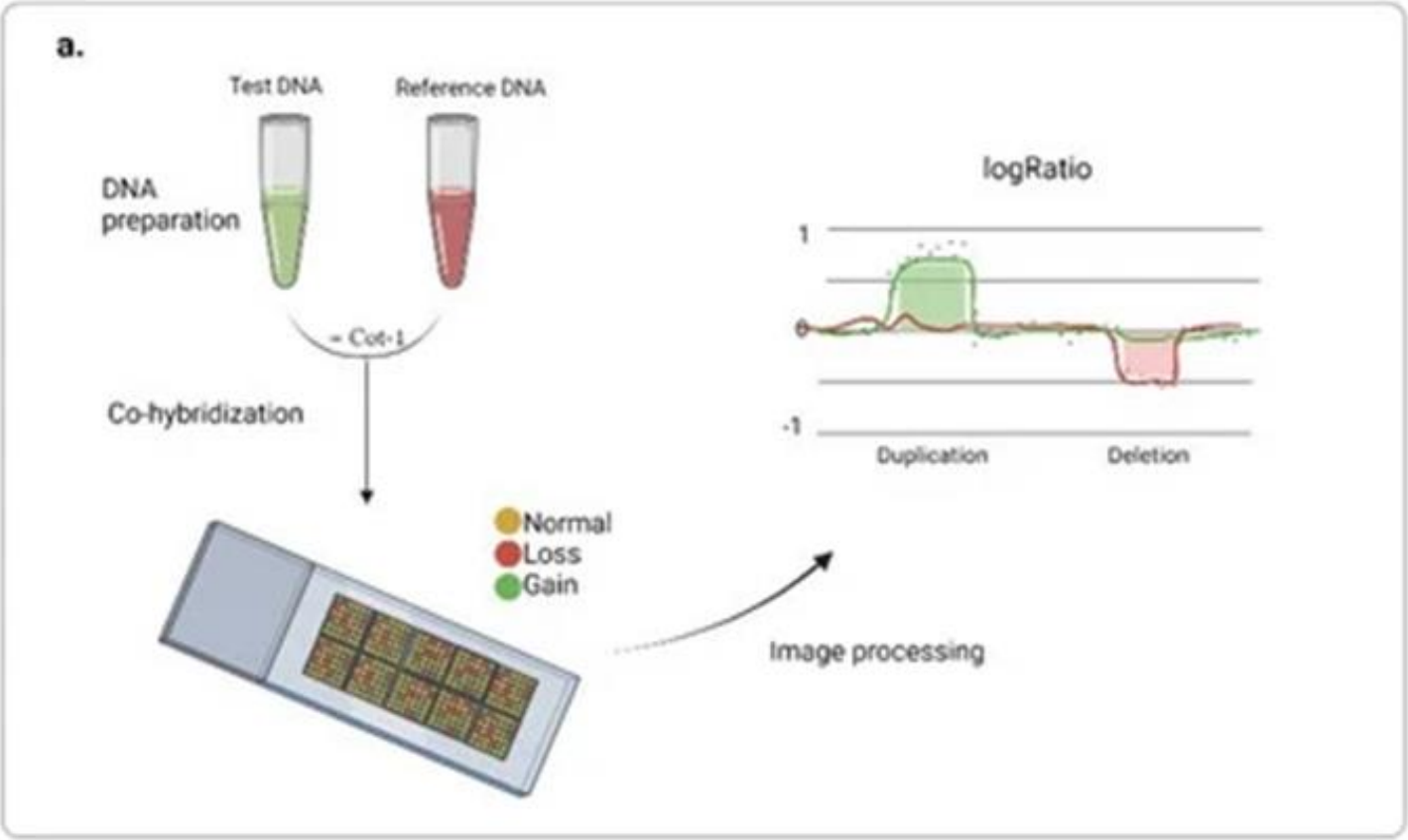
1. Genome-wide single-nucleotide polymorphisms (SNP) arrays which assay SNPs across the genome,

   https://www.youtube.com/watch?v=Naona1y_I2U

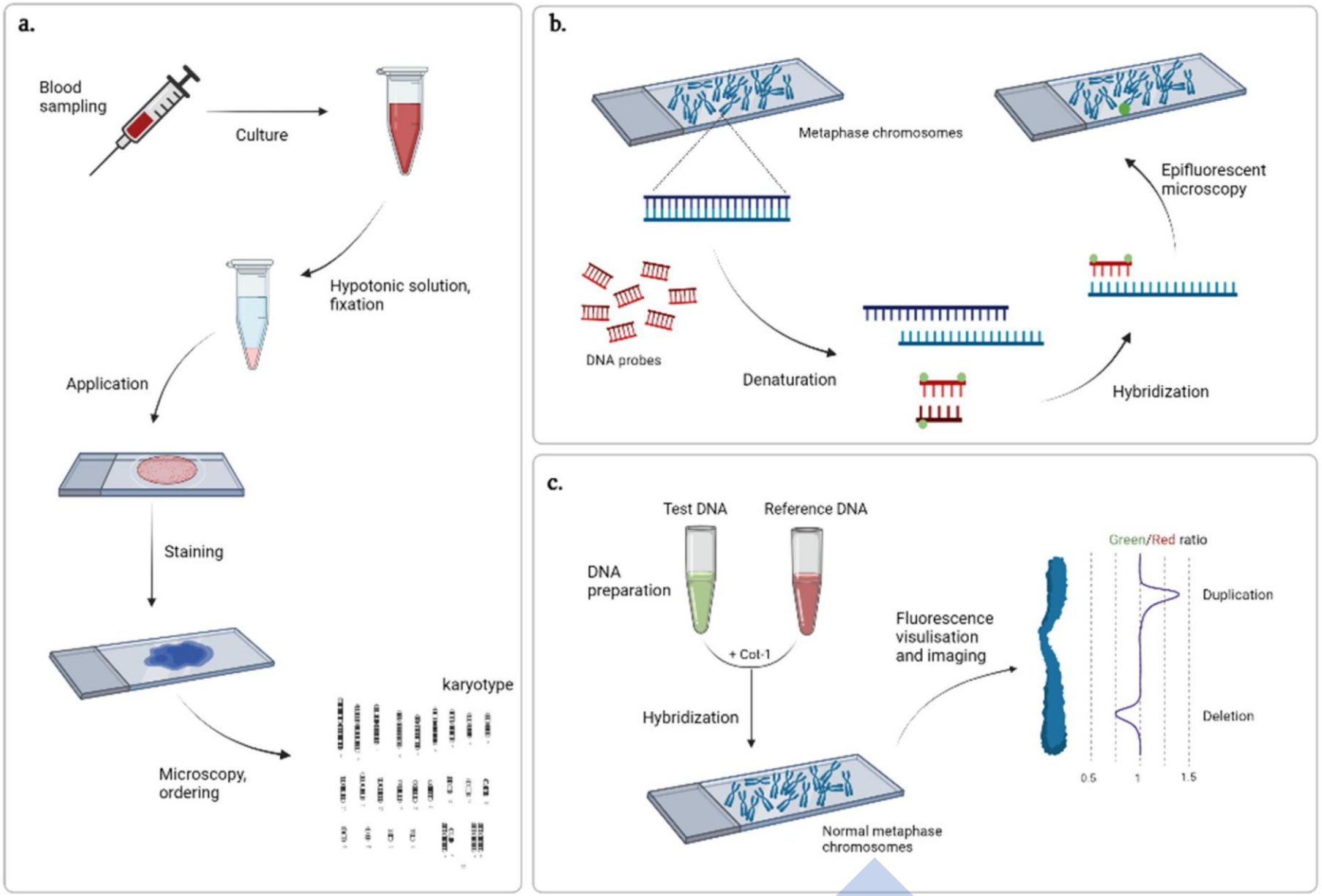
2. Array comparative genomic hybridization (aCGH) which use larger oligonucleotide probes tiled across the genome, often with a “backbone” of SNP probes.

Array-based comparative genomic hybridization (aCGH)

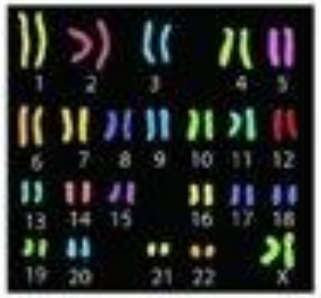
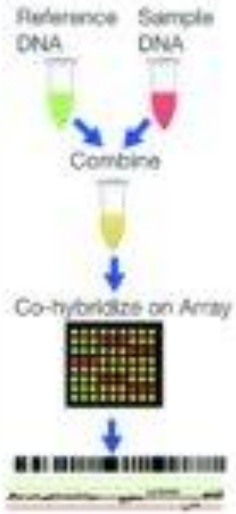
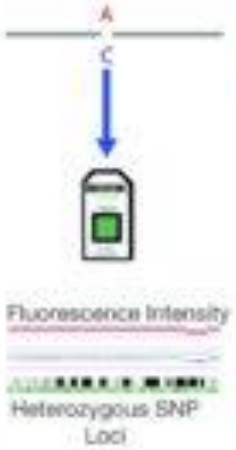
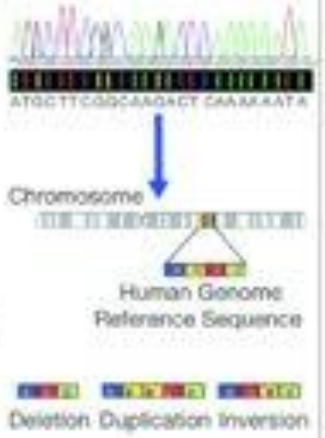
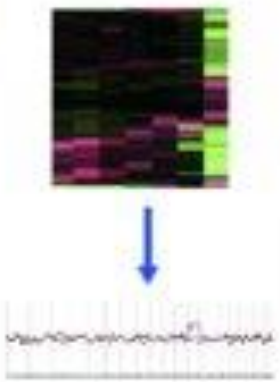
- Hybridization occurs on multiple DNA probes attached to a solid surface, and physical position on a chip and specific nucleotide sequences of the probes are pre-determined, which allows visualization of relative fluorescence intensities of test and control samples along the genome.



Cytogenetic techniques: (a) karyotyping, (b) FISH, and (c) comparative genome hybridization.



A Comparison of Available Methods to Analyze the Genomic Integrity of Stem Cells

	Cytogenetic		DNA-based			RNA-based
Methods	G-banding Karyotyping	Spectral Karyotyping 	aCGH 	SNP Array 	Whole-genome Sequencing 	Expression Array 
Resolution	~10 Mb	Variable (depends on aberration type)	Less than 1 Mb	Less than 1 Mb	Single base-pairs	~10 Mb
Sensitivity	~5%	~5%	Over 20%	Over 20%	Over 20%	Over 30%
Cost	High (hundreds of $)	High (hundreds of $)	High (hundreds of $)	High (hundreds of $)	Very High (thousands of $)	High (hundreds of $)
Unique Strengths	• Easy and reliable • High sensitivity for mosaic cultures • Detection of balanced translocations and inversions	• High sensitivity for mosaic cultures • Detection of balanced translocations and complex karyotypes	High resolution	• High resolution • Detection of LOH	• Ultimate resolution • Detection of point mutations	• Same biological material used for expression profiling and genomic integrity analysis • Identification of involved genes
Unique Limitations	Low resolution	• Low ratio of resolution-per-cost • Inability to identify small duplications and deletions	• Low sensitivity for mosaic cultures • Cannot detect polyploidy	Low sensitivity for mosaic cultures	Very expensive	• Sex chromosomes are not readily analyzed • Low sensitivity for mosaic cultures

Genetic analysis:

- Changes that occur in germ cells (egg or sperm) are inherited by offspring.

- Changes that occur in somatic cells (body cells other than egg or sperm) are not passed to future generations but can result in disease for the individual possessing them (e.g., cancer).

- Changes—sometimes called mutations—can occur as a result of mistakes in coding in the coding nucleotides, rearrangements within the gene, insertion of new genetic material into the gene, or duplication or deletion of parts or all of a gene.

Genetic analysis:

- Disorders resulting from changes in one gene alone are called monogenic (e.g., cystic fibrosis, sickle cell anemia, Duchenne muscular dystrophy).

- Disorders resulting from changes in several genes, usually in combination with an environmental influence, are called multifactorial. Multifactorial disorders (e.g., common types of coronary heart disease and most forms of diabetes) tend to affect far more individuals than do monogenic disorders.

Human monogenic disorders:

- The altered gene can be located on any one of the 22 autosomal chromosomes or on a sex chromosome.
- Modes of inheritance of the altered gene can be autosomal dominant, autosomal recessive, or X-linked.
- In an autosomal dominant disorder, a single abnormal gene causes the trait to be expressed, even though the corresponding gene (all autosomal genes are paired) is normal.
- Thus, an individual with an autosomal dominant disorder will have one mutant gene and one normal gene (unless both parents are affected), and the person will usually be symptomatic at some level, although symptoms might vary in severity and age of onset.

Human monogenic disorders:

- In autosomal dominant disorders, an affected individual generally has an affected parent.

- Each child of an affected individual with an autosomal dominant disorder has a 50 percent chance of inheriting the mutant gene, with the same risks for both males and females. In some cases, however, the mutation arises spontaneously (also called sporadic mutations) and is not inherited from either parent.

- The proportion that is due to a new mutation varies greatly for various disorders and is highest for the most severe diseases; this is because those most severely affected usually do not reproduce.

Autosomal recessive disorders result in illness only if a person receives two copies of the mutant gene, one from each parent who is a carrier.

- Such a person is considered homozygous for the gene and is affected with the disorder.
- Individuals who carry only one copy of the mutated gene are called heterozygotes, or carriers, and are clinically asymptomatic.
- Examples of autosomal recessive disorders are cystic fibrosis (CF), Tay-Sachs disease, phenylketonuria, sickle cell anemia, and thalassemia.

- Autosomal recessive disorders tend to occur with varying frequencies among different racial and ethnic populations.

- If both parents are carriers of the same recessive trait, each pregnancy carries a 1 in 4 risk of producing an affected child (homozygous affected) and a 1 in 2 chance of producing an asymptomatic carrier, such as themselves. A 1 in 4 chance also occurs with each pregnancy that the child with be neither a carrier nor affected by the disease.

Sex chromosomes:

- Each individual has two sex chromosomes in the normal condition; males have an X and a Y chromosome, females have two X chromosomes.

- Theoretically, altered genes can occur on either the X or the Y chromosome, but the Y chromosome carries few genes.

- In X chromosomal recessive disorders (X-linked), males are disproportionately affected by these disorders because they possess only one X chromosome, which carries the mutation.

Sex chromosomes:

- Carrier females have one normal chromosome and one carrying the altered gene.
- Recessive diseases caused by alterations in genes on the X chromosome include Duchenne muscular dystrophy, fragile X syndrome, and hemophilia.
- In X-linked dominant diseases, men and women are affected equally since the abnormal gene dominates its normal partner; there can never be male to male transmission, however, because an affected father only passes a Y chromosome to his sons.

Heterogeneity:

- A phenomenon called heterogeneity sometimes adds to the complexity of determining inheritance within and across families.

- For example, different genes located on different chromosomes can independently give rise to the clinically identical phenotype, as is the case for retinitis pigmentosa, an inherited form of blindness.

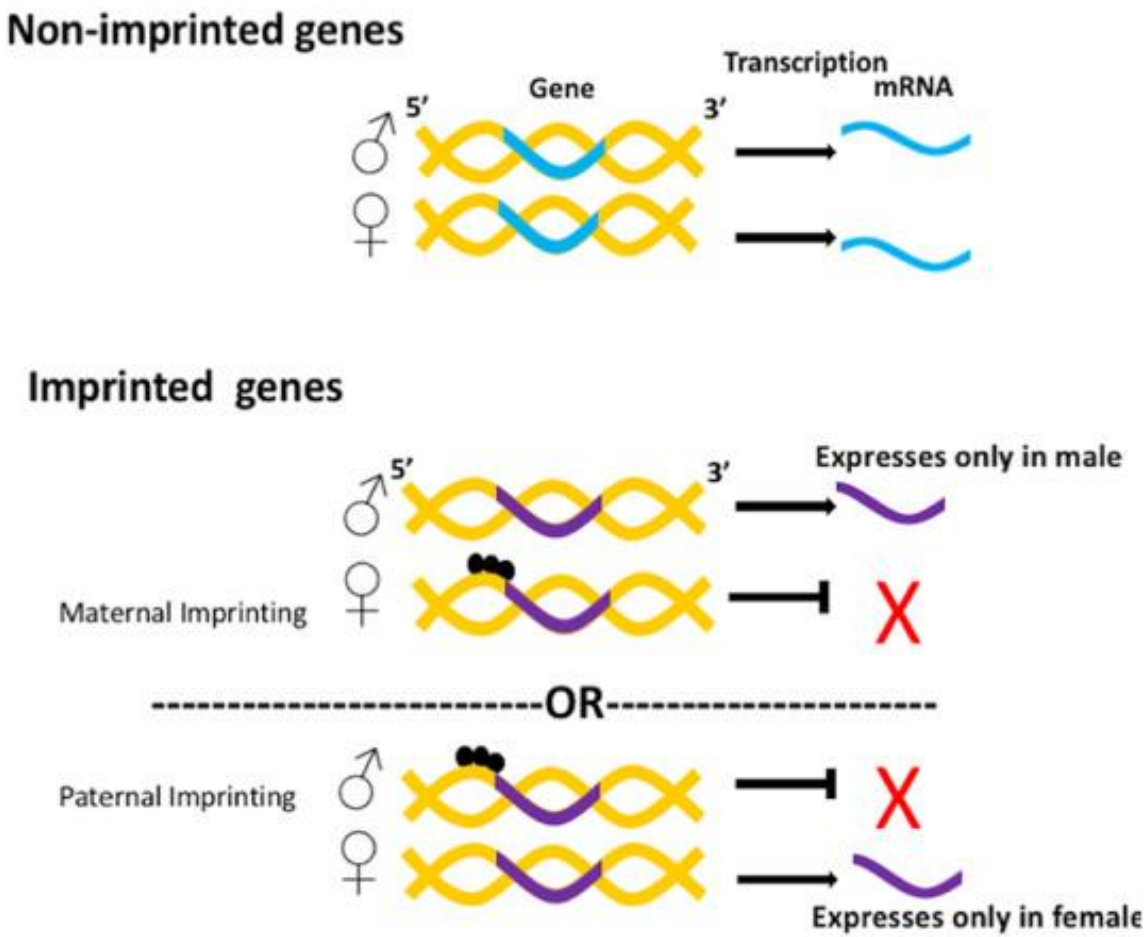
- Another form of heterogeneity, allelic heterogeneity, is caused by different mutations in a single gene that give rise to manifestations of a disease.

How many mutations can cause CF?

- Genetic testing identifies the most common mutations. Scientists have found more than 1,700 different mutations in the CFTR gene that can cause CF.

Genomic imprinting:

Genomic imprinting is an epigenetic phenomenon that causes genes to be expressed or not, depending on whether they are inherited from the mother or the father. Genes can also be partially imprinted.



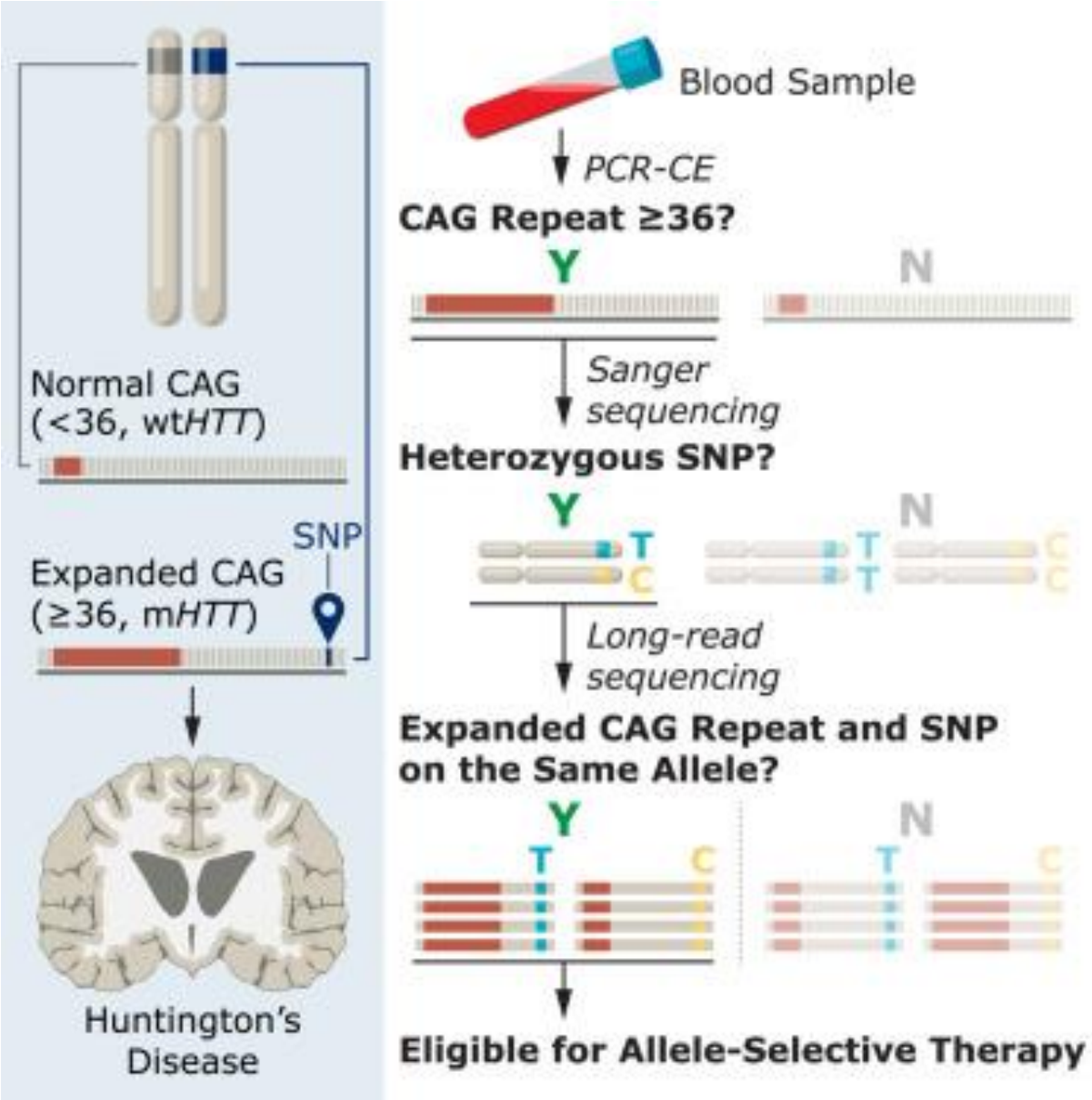
Mitochondrial disorders:

- Mitochondria are energy-generating organelles (components inside the cytoplasm of every cell) and carry their own chromosomal DNA.
- They are thought to be descendants of ancient bacteria that migrated into animal cells and became essential functional parts of those primitive cells.
- Sperm lose their mitochondria when they penetrate the egg; thus, mitochondrial genes (and any mutations in them) are passed only from the mother.
- Mutations in mitochondrial genes can be detected by molecular techniques and have been positively associated with certain types of blindness (Leber optic atrophy), muscle diseases, a type of epilepsy, and dementias associated with aging.

Allelic expansion:

- Here, instead of DNA remaining constant over the generations, as is usually the case, a gene segment expands in size when transmitted from parent to child.

- The expanded gene may cause a characteristic disease such as Huntington disease, myotonic dystrophy, spinal bulbar muscular atrophy, or fragile X mental retardation syndrome.



Technologies for Detecting Genetic Disorder

- Early stages, observation of an individual's clinical findings or constellation of anomalies, and on an assessment of the family history.
- Later, biochemical assays were developed to test for inborn errors of metabolism, such as phenylketonuria or sickle cell disease.
- These earlier techniques remain useful today.
- Chromosomal analysis has been in use for more than 30 years to diagnose errors in number or shape of chromosomes that can result in genetic disorders and disease.

DNA diagnosis

- Two major approaches are currently used in DNA diagnosis and study of genetic disease—direct tests and indirect tests (known as linkage analysis).
- In direct analysis of the gene or gene variant, family members need not be tested.
- Direct tests are now being conducted for cystic fibrosis, the thalassemias, and Duchenne muscular dystrophy.
- In disorders like CF, for which more than 1700 mutations have been identified, screening for all mutations is not feasible. In most cases, only the more common mutations are searched for in testing.

DNA diagnosis

- The second approach—linkage analysis—is used in cases where the gene has been localized to a region of a chromosome, but the gene has not been cloned; tracking the inheritance of linked DNA markers provides a means of predictive testing.

- In this situation, DNA from family members, including at least one affected family member, is essential to the determination of the genetic status of an individual.

What is linkage analysis used for?

- Linkage analysis: Study aimed at establishing linkage between genes.
- Today linkage analysis serves as a way of gene-hunting and genetic testing.
- Linkage is the tendency for genes and other genetic markers to be inherited together because of their location near one another on the same chromosome.

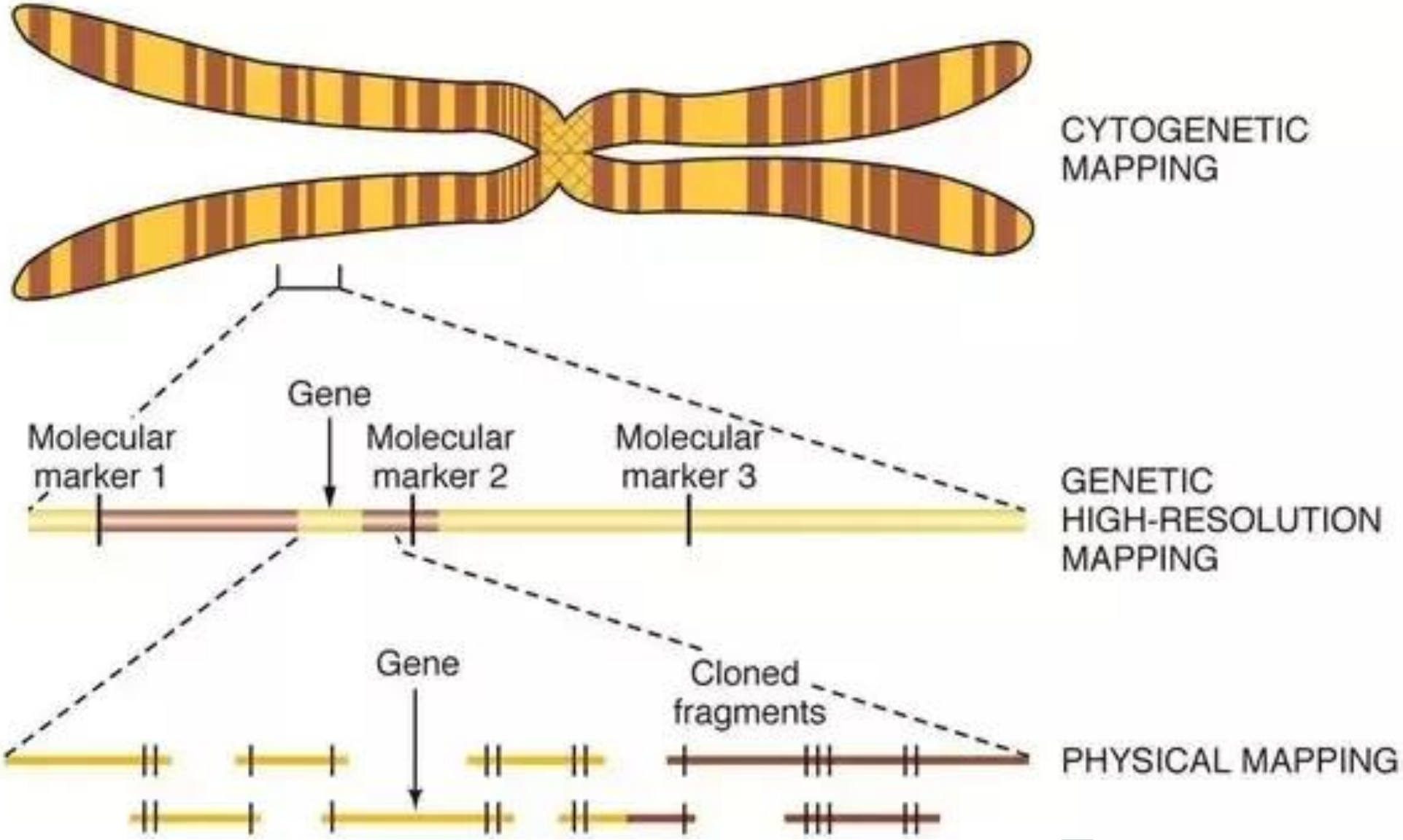
What does linkage analysis determine?

- A gene-hunting technique that traces patterns of disease in high-risk families. It attempts to locate a disease-causing gene by identifying genetic markers of known chromosomal location that are co-inherited with the trait of interest.

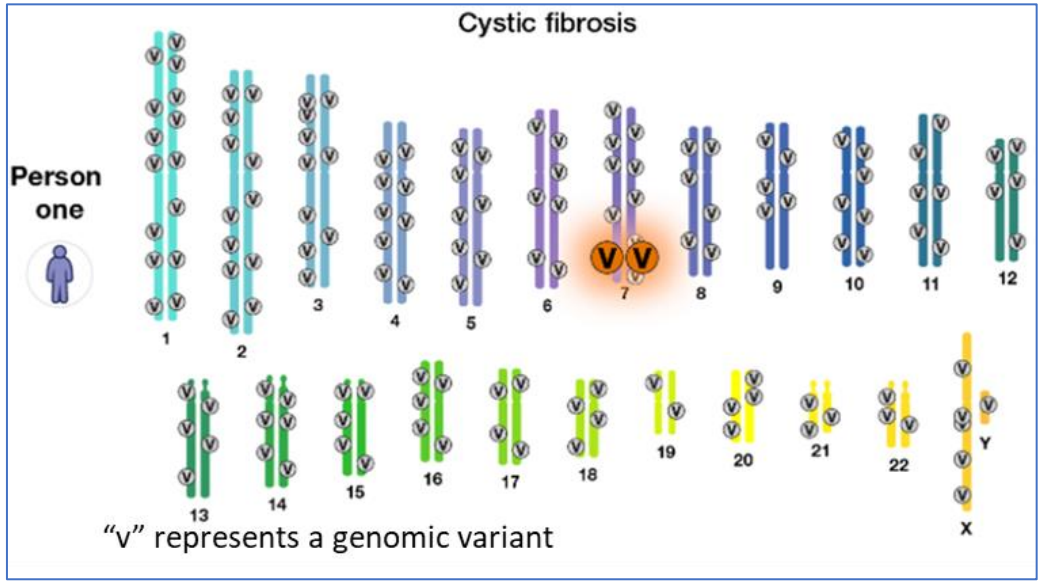
What is the basic principle of genetic linkage analysis?

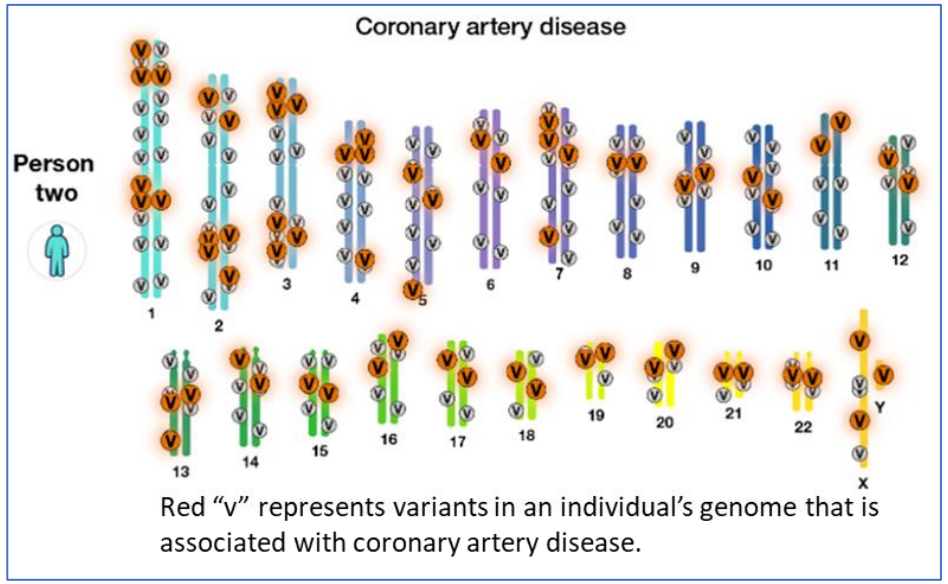
- The fundamental principle of linkage analysis is that for any two loci on the same chromosome, the closer they are to each other, the less likely it is that they will undergo recombination.

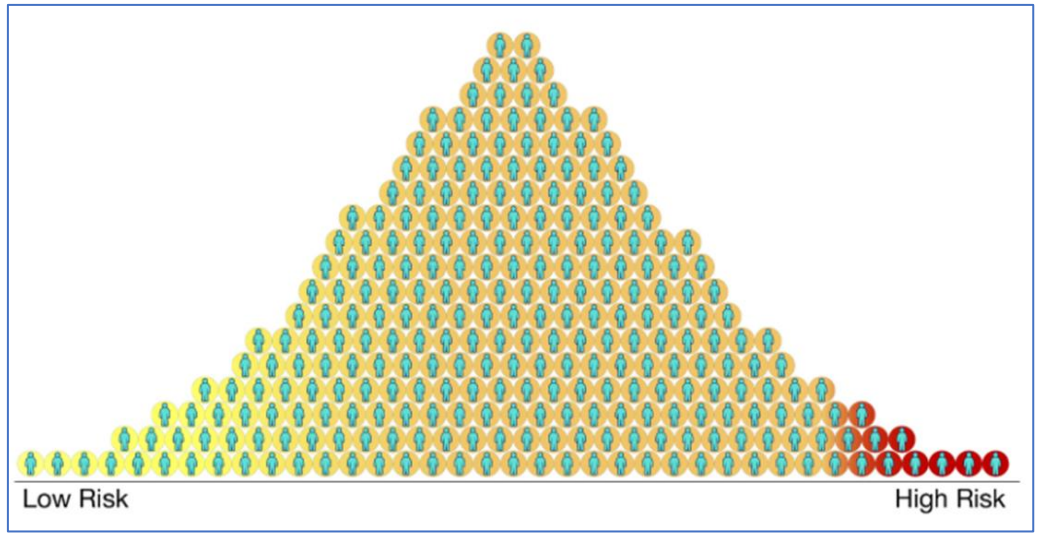
- A phenomenon known as crossing over can occur between the marker and the gene, which can lead to erroneous diagnostic conclusions.


CYTOGENETIC MAPPING
Gene
Molecular marker 1
Molecular marker 2
Molecular marker 3
GENETIC HIGH-RESOLUTION MAPPING
Gene
Cloned fragments
PHYSICAL MAPPING

Polygenic risk scores







Each polygenic risk score can be put on a bell curve distribution. Most people will find their scores to be in the middle, indicating average risk for developing a disease. Others may find themselves on the tail ends, putting them at either low or high risk. People with scores on the high-risk portion of the spectrum may benefit from discussions about this risk with their physicians and genetic counselors for further health assessments.

What are the types of genetic markers?

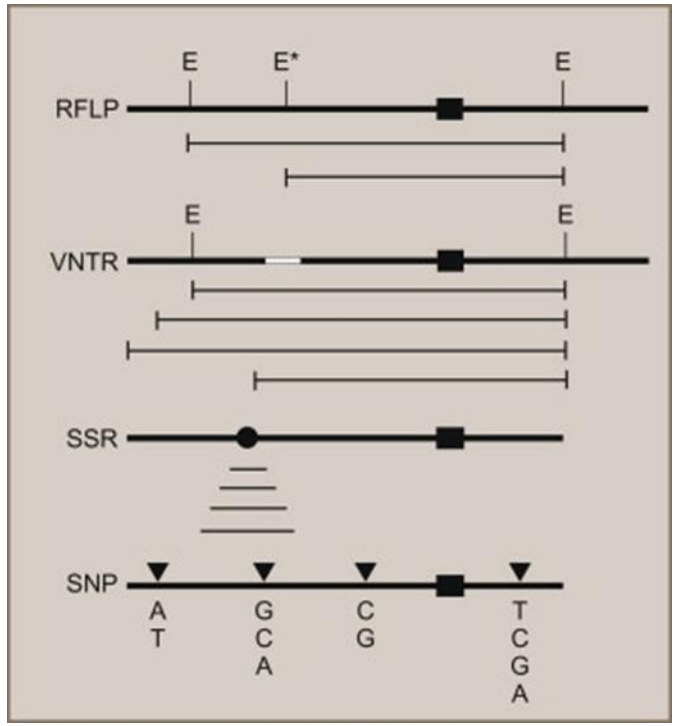
- In the human genome, the most common types of markers are RFLP, VNTR, SSR, STR, SNP
- DNA polymorphisms are produced by changes in the nucleotide sequence or length.
- **These result from:**
    I. Variations in the fragment length pattern produced after digesting DNA with restriction enzymes,

    II. Variations in the size of a DNA fragment after PCR amplification, and

    III. Variations in the DNA sequence itself.

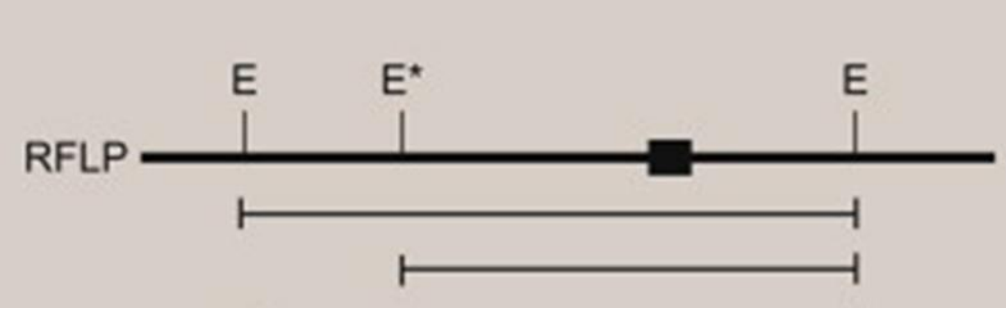
DNA polymorphisms include:

1. RFLP – restriction fragment length polymorphism;
2. VNTR – variable number of tandem repeats;
3. SSR – simple sequence repeats or STR – simple tandem repeat, i.e. microsatellites, and
4. SNP – single nucleotide polymorphism.



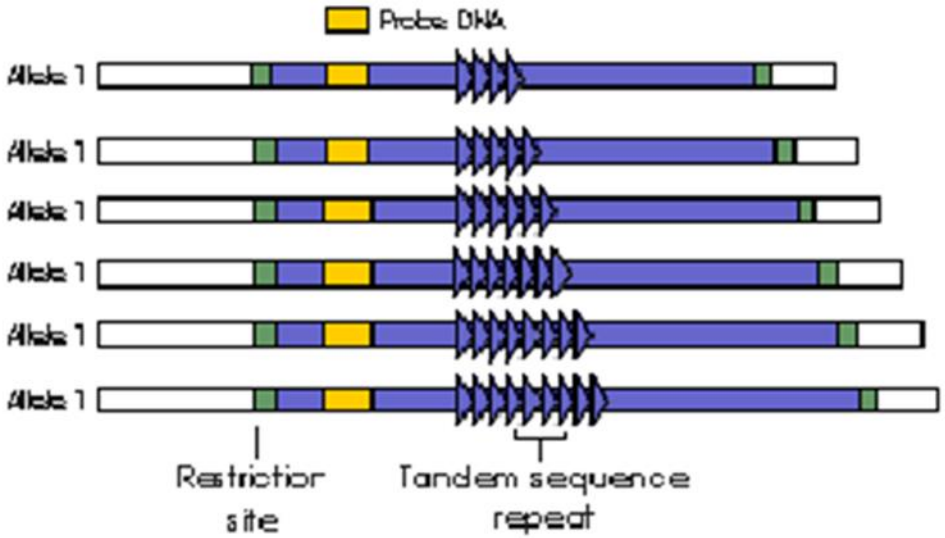
RFLP: A segment of DNA is digested with a restriction enzyme E.

- This segment can be identified in Southern blot analysis by using a DNA probe that will hybridize to the segment marked (■) or PCR can be used to amplify this specific region.
- RFLPs result from point mutations affecting a single restriction enzyme recognition site (E*) which will either be absent or present.
- If absent, enzyme E will digest the DNA at the two outside E sites; if present, enzyme E will digest DNA at E* as well as the two outer E sites.
- The position of the probe means only fragments generated from E* and the E site on the right will be detected.
- Therefore, RFLPs are biallelic, i.e. they give two fragment sizes (large or small) depending on whether the polymorphic restriction fragment site (*) is absent or present respectively.



VNTR: The multiallelic VNTR has the potential to be more polymorphic (and so more informative) since the changes in the E-specific restriction fragment are brought about by the insertion of a variable number of repeat units at the polymorphic site (hatched area).

- Thus, more polymorphic DNA fragments are generated, e.g. the four different sized fragments illustrated.
- Because of their greater intrinsic variability, VNTRs are more informative since there is a greater chance that heterozygous patterns will be detected at any one locus.
- Examples of VNTRs detected by restriction enzyme digests are the minisatellites.

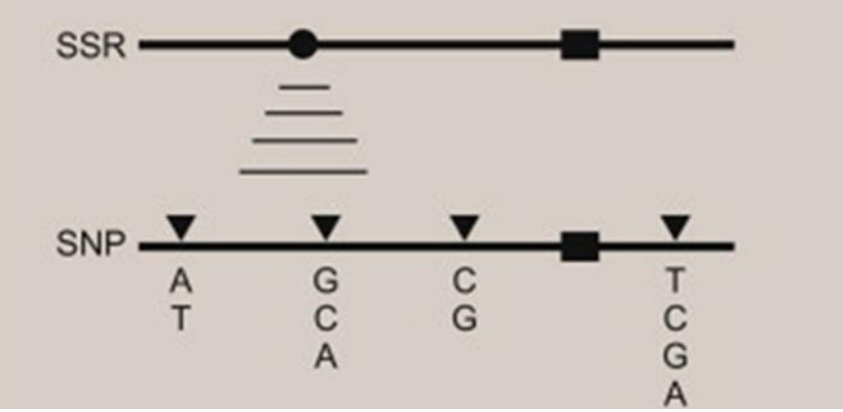


SSR: In contrast to RFLPs or VNTRs that can be identified by Southern analysis and PCR, SSRs (or STRs) are much smaller in size and so are detectable only by PCR.

- They are the microsatellites and are polymorphic because repeats involve simple sequences (●) such as a (CA)n where n is a number usually between 2–4.
- Amplification of DNA containing a SSR will produce fragments of variable size.

SNPs: These single base changes are similar to RFLPs but SNPs are more useful because they do not have to change the restriction enzyme digestion pattern to be detected by DNA sequencing.

- SNPs can be biallelic or have more than two alleles.
- Because these are found frequently throughout the genome (four are depicted here in relationship to ■ which could represent a gene) they have the potential to discriminate alleles more effectively.



Basic human genetics and genetic analysis

- Newborn screening
- Carrier testing and screening (We are all potential carriers of several recessive genes that would prove lethal to our offspring if they received a double dose)
- Prenatal diagnosis
- Testing for late-onset disorders

Genetic Testing and Assessment:

When we can do the genetic testing for genetic diseases?

- Identifying a specific mutation,
- One can do so in gametes,
- In the zygote immediately after conception,
- In the early embryo,
- Prenatally throughout pregnancy,
- In the newborn period,
- In childhood or adolescence,
- As part of reproductive planning in adulthood, or thereafter.

- Assessment of the molecular basis for inheritance is done by examining the specific structure and function of genetic material, or DNA. Locating a disease-causing gene on a chromosome.

Technologies for genetic testing:

- Chromosomal,
- Biochemical,
- DNA-based techniques.

Samples for testing;

- Blood,
- Amniotic fluid,
- or other tissue.
- In addition, DNA for linked marker gene.

- The purpose of the following sections is to illustrate how genetic tests are used in practice and to identify issues raised by their use in various types of testing and in various populations.

Cytogenetics:

- Study of the number of and structure of the chromosomes through microscopic analysis.

- It can provide information about the person's gender and about certain diseases that are directly related to abnormal numbers or configurations of the 23 pairs of chromosomes found in humans (e.g., Down syndrome, fragile X syndrome).

- Locating a disease-causing gene on a chromosome and isolating it are an important goal of research.
- The gene's structure and function may provide opportunities for diagnosis and may lead to treatment of the disorder.

- Understanding gene expression and its regulation is the key to understanding genetic disease and hereditary variation. Hereditary variation is the result of changes—or mutations—in DNA.

There are three major methods of cytogenetic testing:

- Routine karyotyping.
- Fluorescent in situ hybridisation (FISH) Comparative genomic hybridisation (CGH) and
- Array comparative genomic hybridisation (aCGH).

Karyotype is a test to identify and evaluate the size, shape, and number of chromosomes in a sample of body cells. Extra or missing chromosomes, or abnormal positions of chromosome pieces, can cause problems with a person's growth, development, and body functions.

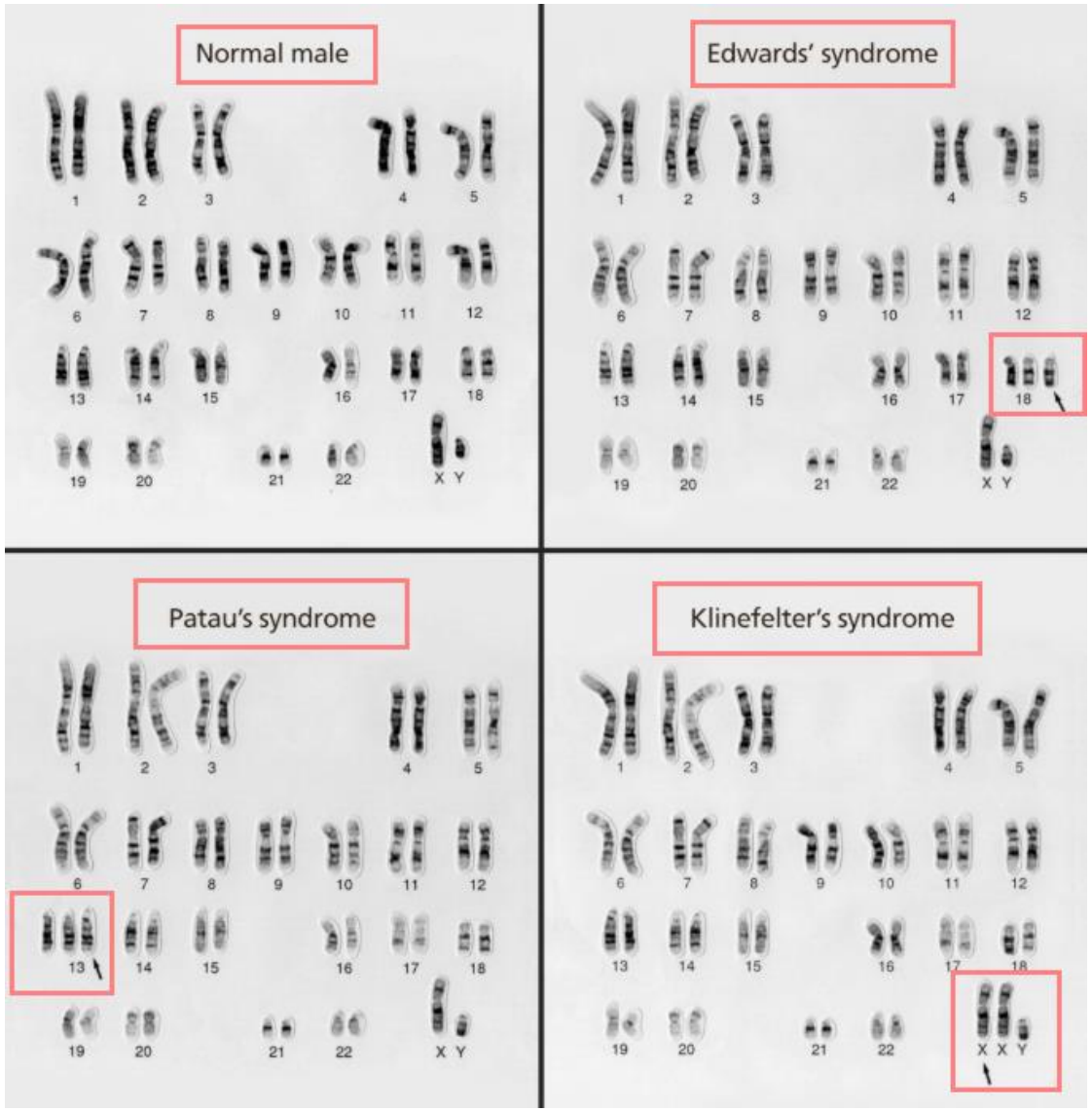
- A karyotype is a representation of the person’s chromosomes. To get a picture of chromosomes, the chromosomes are isolated, stained, and checked under the microscope.

- Through the microscopic examination, a picture of the chromosomes is taken, cut up, and rearranged according to their size from largest to smallest.

Karyotyping procedure in animation

https://www.youtube.com/watch?v=bx2g6ywFCk8

Karyotyping



Chromosomal abnormalities that are detected using karyotyping are categorized into the following:

1. Trisomies – There are three copies of one of the chromosomes. In a normal circumstance, there should only be two copies of a chromosome.

2. Monosomies – Instead of two copies of a chromosome, there is only one copy.

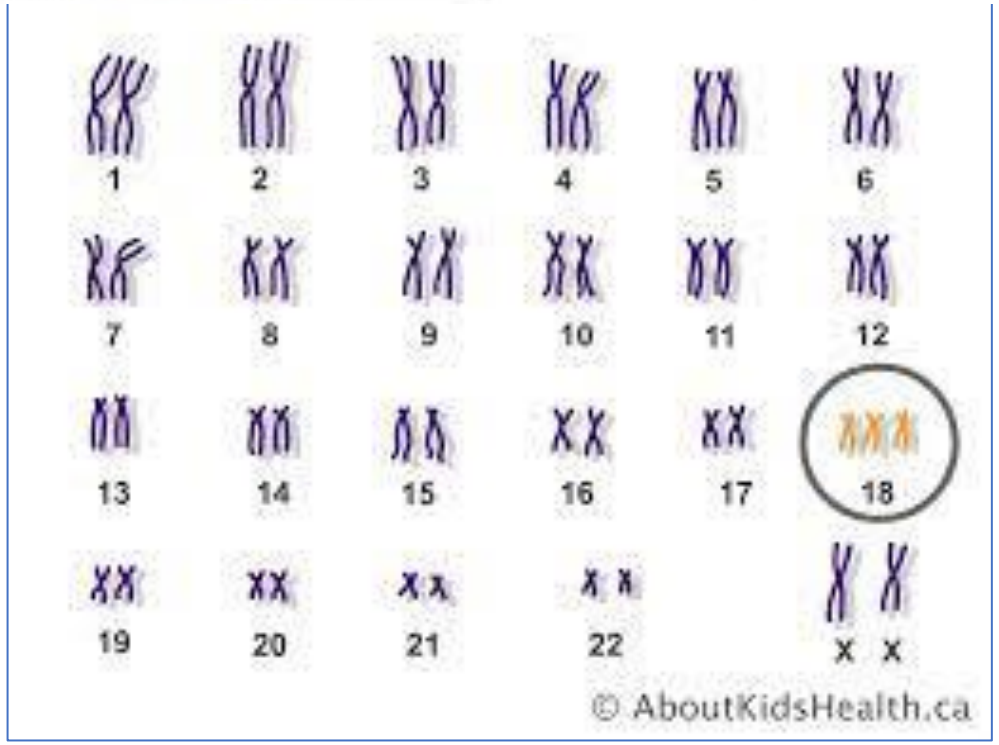
3. Chromosome deletions – A part of the chromosome is missing.

4. Chromosomal translocations – A part of the chromosome attaches to another chromosome and vice versa.

Trisomy 18 (Edwards syndrome)

- Trisomy 18 is a rare condition caused by an extra copy of chromosome 18.
- Children with trisomy 18 have severe developmental delay, as well as severe birth defects and health problems involving nearly every organ system in the body.
- It is difficult to predict the life expectancy of a baby with trisomy 18.
- Parents of a baby with trisomy 18 are encouraged to seek genetic counselling.

- Congenital heart defects
- Growth retardation
- Dysmorphic features (see diagram)
- Facial clefts
- Spina bifida
- Sever developmental delay



Diagnosis of trisomy 18

- Babies with trisomy 18 have a unique group of characteristics and can be diagnosed by physical examination.
- To confirm the diagnosis, a small blood sample can be taken and the chromosomes can be analyzed to determine the presence of an extra chromosome 18.

- Chromosomal abnormalities can also be diagnosed before birth by analyzing cells in the amniotic fluid obtained by amniocentesis, or from the placenta, obtained by chorionic villus sampling (CVS) or from the fetal blood obtained by cordocentesis.

- Screening for trisomy 18 is also available through first trimester screening (FTS), integrated prenatal screening (IPS) or by non-invasive prenatal testing (NIPT).

Diagnosis of trisomy 18

- The diagnosis of trisomy 18 can also be suggested by detailed fetal ultrasound; however, ultrasound is not 100% accurate, since not all abnormalities can be seen on ultrasound and the same abnormalities seen prenatally in trisomy 18 can also be seen in other conditions. In contrast, a chromosome analysis, whether performed on fetal/newborn blood sample, cells from the amniotic fluid or placental sample, is over 99.9% accurate.

Life expectancy of a baby with trisomy 18

- Fifty per cent of babies born with trisomy 18 survive beyond their first six to nine days.
- About 12% of babies born with trisomy 18 survive the first year of life.

- It is difficult to predict the life expectancy of a baby with trisomy 18 if the baby does not have any immediate life-threatening problems. For babies that have survived their first 30 days of life, 36% were alive at one year. About 10% of children born with trisomy 18 survive until 10 years of age.

- There is a small chance of having another child with trisomy 18
- The risk of having a baby with trisomy 18 increases with the mother's age.

- However, the average age of the mother at delivery of a baby with trisomy 18 is 32 years. In general, in each subsequent pregnancy, the chance of having another baby with trisomy 18 is no greater than 1%.

Genetic counselling is recommended

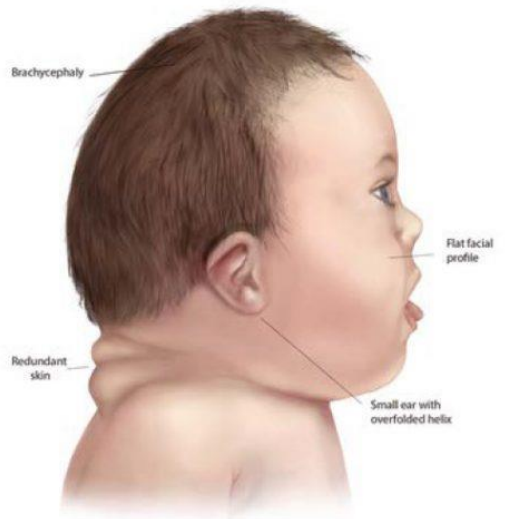
- Parents of a baby with trisomy 18 are encouraged to seek genetic counselling. This can help parents to:
- understand the results of chromosomal tests in detail
- understand the chances that other babies will be affected
- understand the diagnosis options for other pregnancies

Down Syndrome (Trisomy 21) in Children

What are the symptoms of Down syndrome in a child?

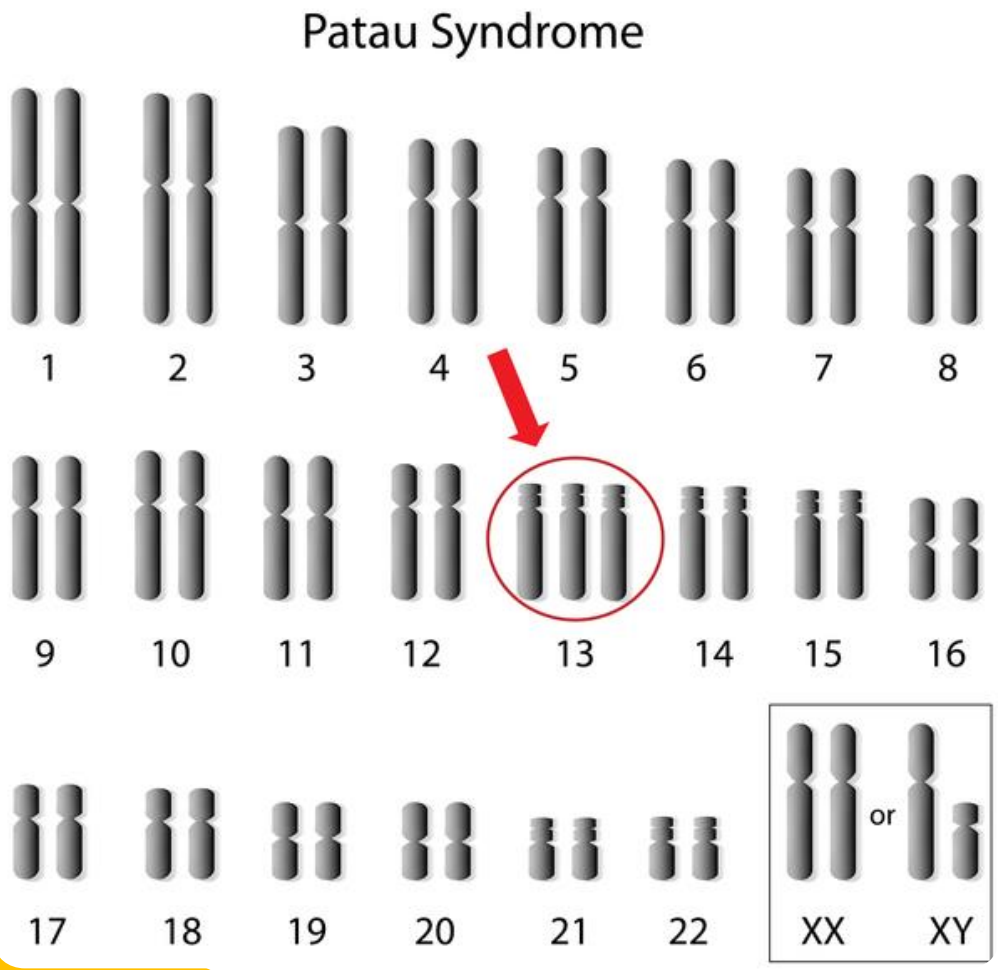
Symptoms can occur a bit differently in each child. They can include:

- Eyes that slant upward
- Small ears that may fold over slightly at the top
- Small mouth that makes the tongue appear large
- Small nose with a flattened bridge
- Short neck
- Small hands with short fingers
- 2 instead of 3 palm creases, including one across the palm and one around the base of the thumb
- Short height
- Loose joints



Chromosome problems such as Down syndrome can often be diagnosed before birth.

- This is done by looking at cells in the amniotic fluid or from the placenta.
- This can also be done by looking at the amount of the baby’s DNA in the mother's blood. This is a noninvasive prenatal screening. These tests are very accurate.
- Fetal ultrasound during pregnancy can also show the possibility of Down syndrome.
- But ultrasound is not 100% accurate. Problems from Down syndrome may not be seen with ultrasound.

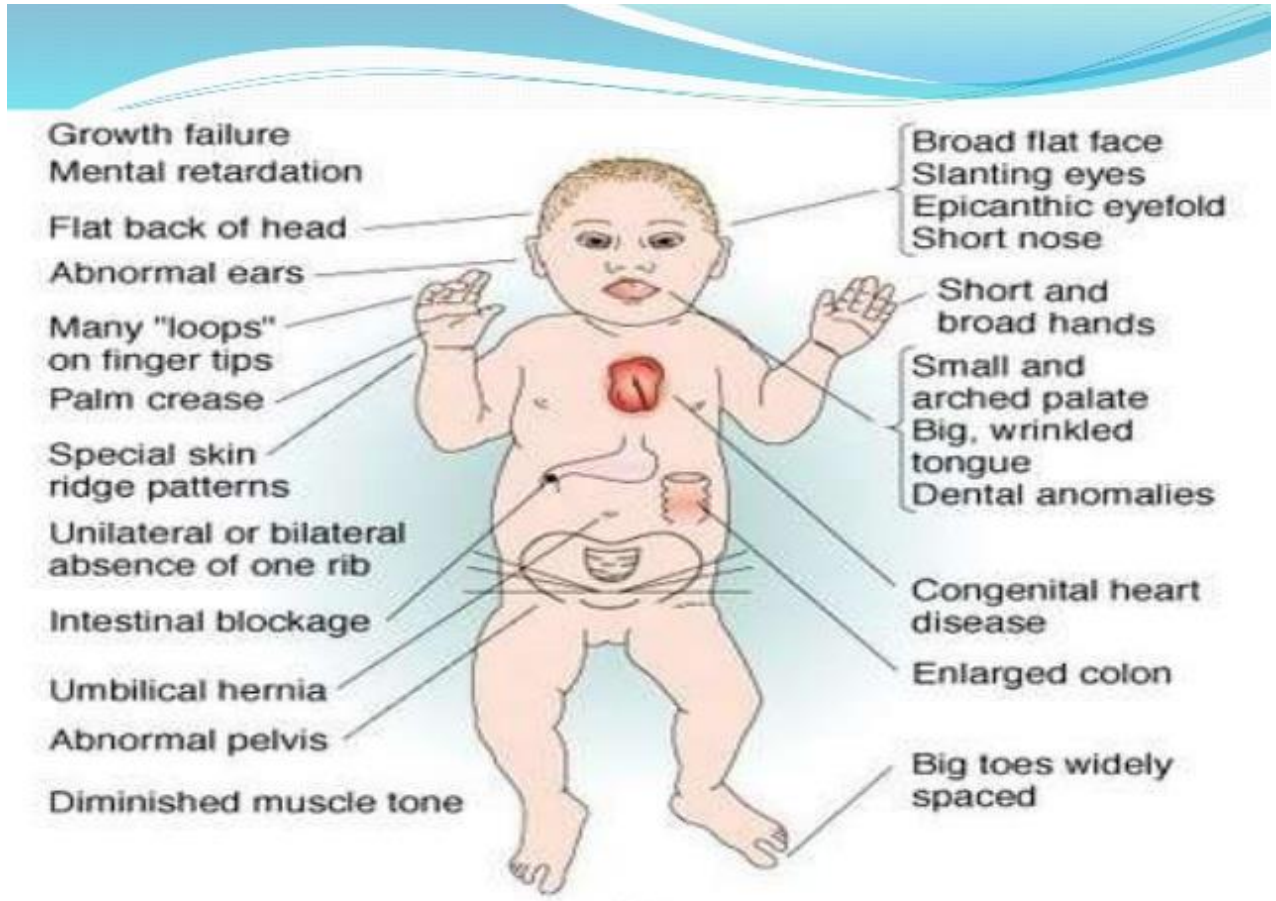


- Most cases of trisomy 13 result from having three copies of chromosome 13 in each cell in the body instead of the usual two copies. The extra genetic material disrupts the normal course of development, causing the characteristic features of trisomy 13.

- Trisomy 13 occurs in about 1 in 16,000 newborns. Although women of any age can have a child with trisomy 13, the chance of having a child with this condition increases as a woman gets older.

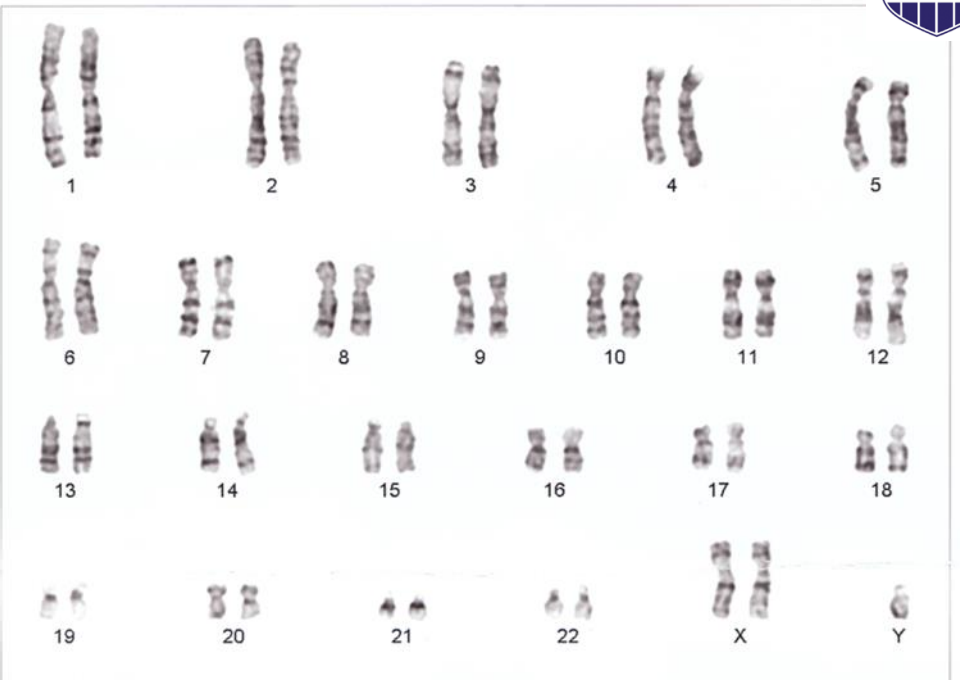
- **Patau syndrome** is a syndrome caused by a chromosomal abnormality, in which some or all of the cells of the body contain extra genetic material from chromosome 13.

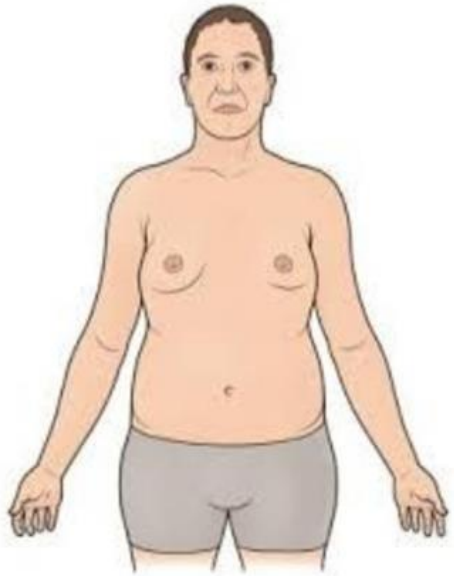
- The extra genetic material disrupts normal development, causing multiple and complex organ defects.



Klinefelter syndrome (KS), also known as 47,XXY, is a syndrome where a male has an additional copy of the X chromosome.

- The primary features are infertility and small, poorly functioning testicles. Often, symptoms are subtle and subjects do not realize they are affected.
- Klinefelter syndrome occurs as a result of a random error that causes a male to be born with an extra sex chromosome. It isn't an inherited condition.
- Humans have 46 chromosomes, including two sex chromosomes that determine a person's sex. Females have two X sex chromosomes (XX). Males have an X and a Y sex chromosome (XY).





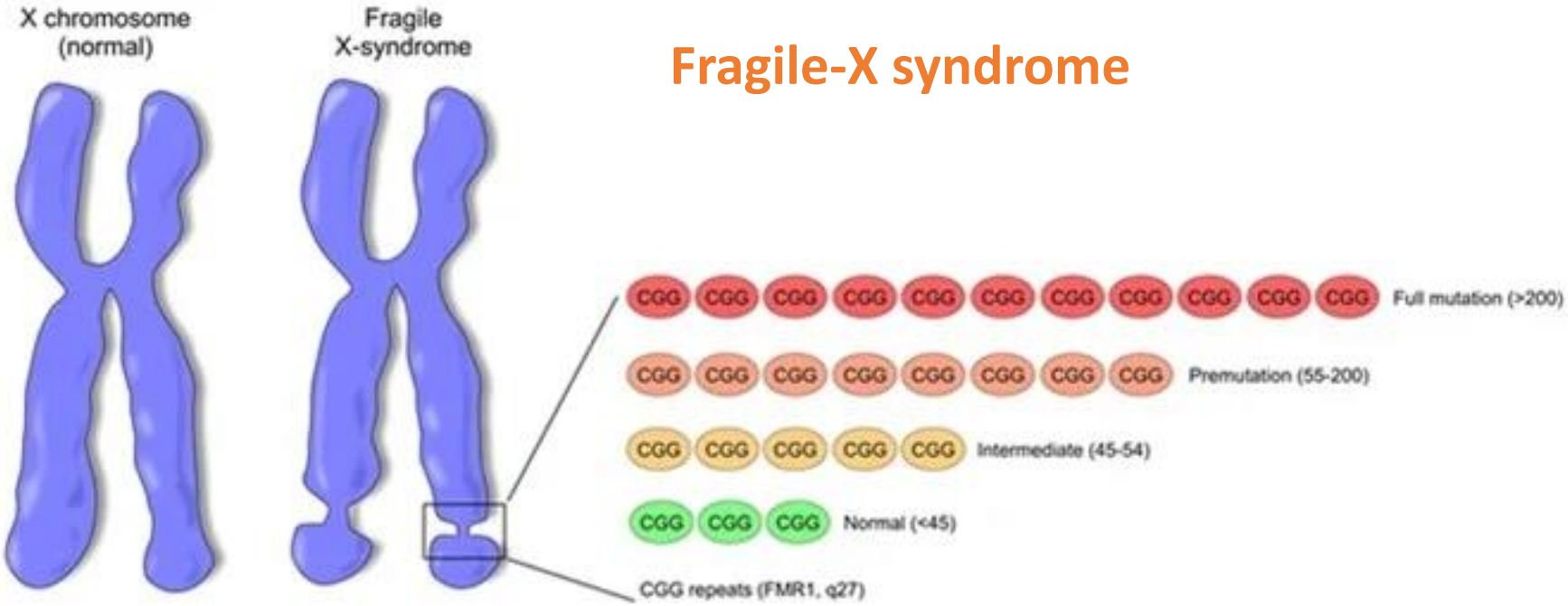
- A taller less muscular body than males there age.
- Broader hips and longer legs.
- Larger breast.
- Weaker bones.
- A lower energy level.
- Smaller penis and testicles
- Delay in puberty or go a parcel amount.
- Less facial and body hair following puberty.

Fragile-X syndrome-Diagnosis

Testing for is recommended if:

- one or more of the disease symptoms are evident in a child
- the mother is known to carry a pre- or full mutation in the FMR1 gene. A pre-mutation is a FMR1 gene with 55 to 200 repeats of the CGG triplet.
- Women with the pre-mutation have an increased risk of having a child with Fragile-X syndrome.
- if there is a history of unexplained mental retardation in the family.

Fragile-X syndrome



World statistics

Fragile X premutation (number of carriers)

More common in men
(approx. 1 in 468 men, 1 in 151 women)

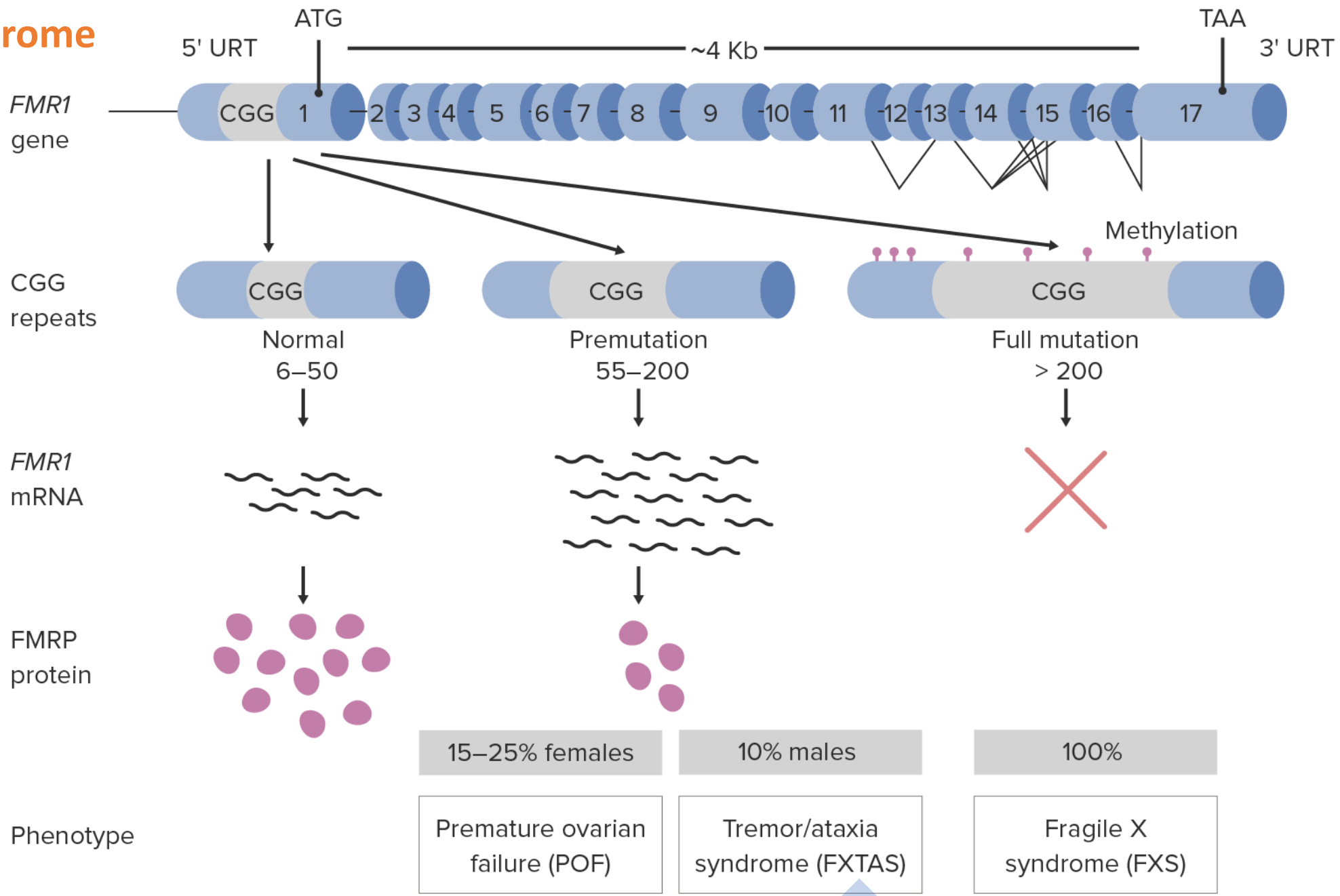
Fragile X full mutation (number of carriers)

approx. 1 in 4000 men, 1 in 6000 women

FXS symptoms and physical tracts

- Long face, Large jaw
- Large, prominent ears
- High arched palate
- Macro-orchidism, macrocephaly
- Joint laxity
- Intellectual disability
- Autism
- ADHD
- Prader-Willi phenotype

Fragile-X syndrome



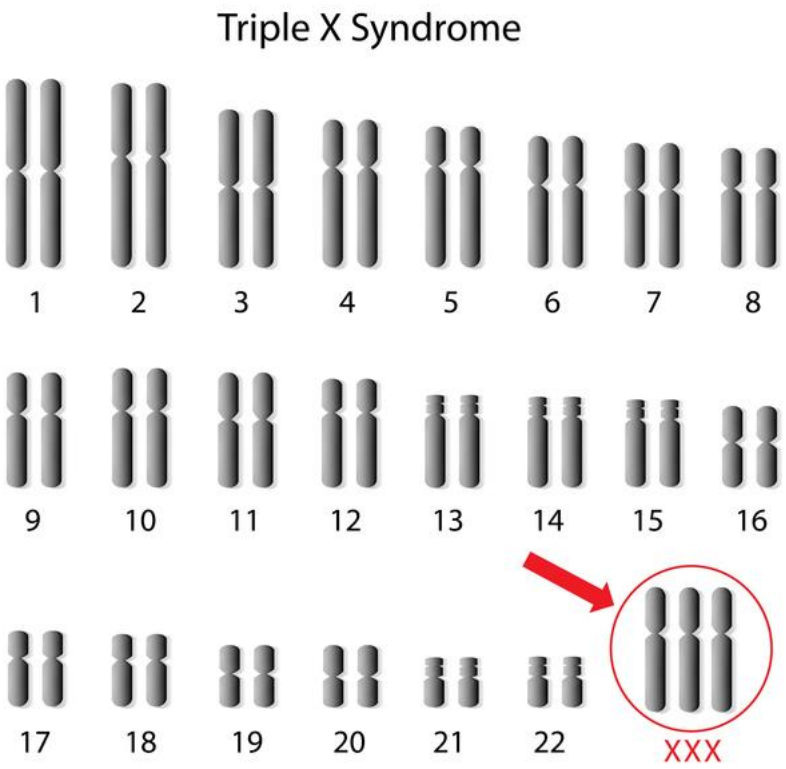
Fragile-X syndrome:

Diagnostic methods include PCR amplification and Southern blotting, which are performed on DNA isolated from peripheral leukocytes.

- Recently, varying immunocytochemical tests have been described to identify fragile X patients, based on the detection of FMR1 protein in cells by a monoclonal antibody.

- This review provides an update on the different DNA methods and gives specific attention to both the newly developed PCR method and antibody methods for prenatal and postnatal diagnosis of the fragile X syndrome.

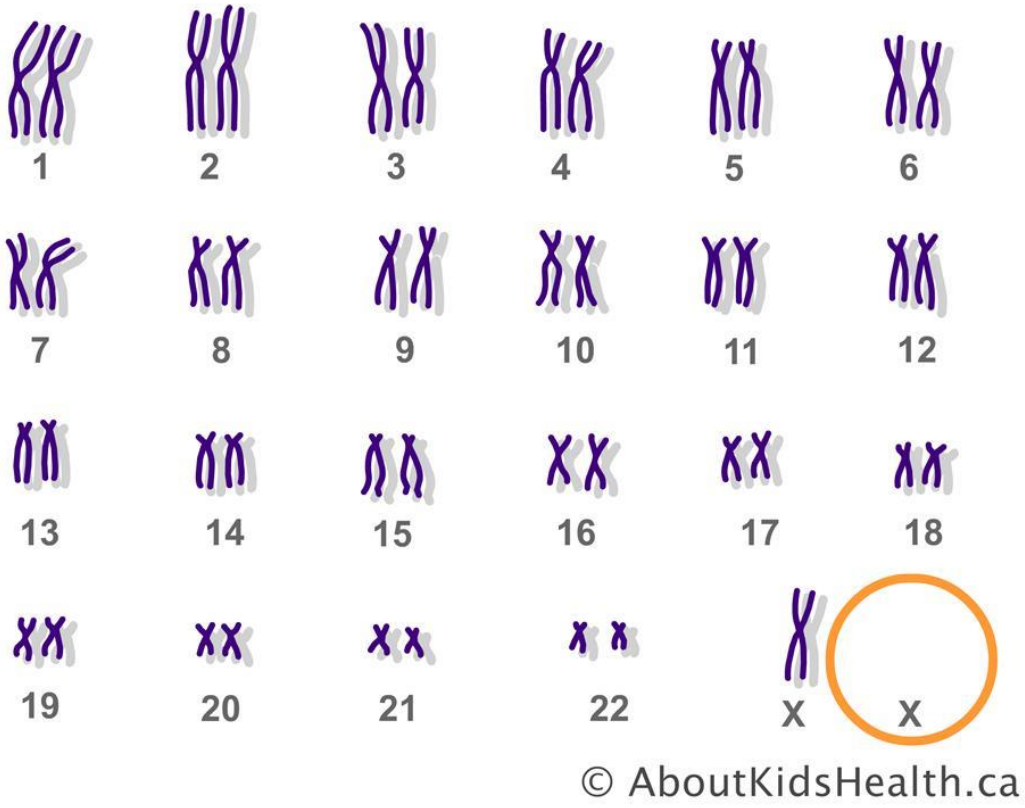
Triple X syndrome, also called trisomy X or 47,XXX, is a genetic disorder that affects about 1 in 1,000 females. Females normally have two X chromosomes in all cells — one X chromosome from each parent. In triple X syndrome, a female has three X chromosomes.



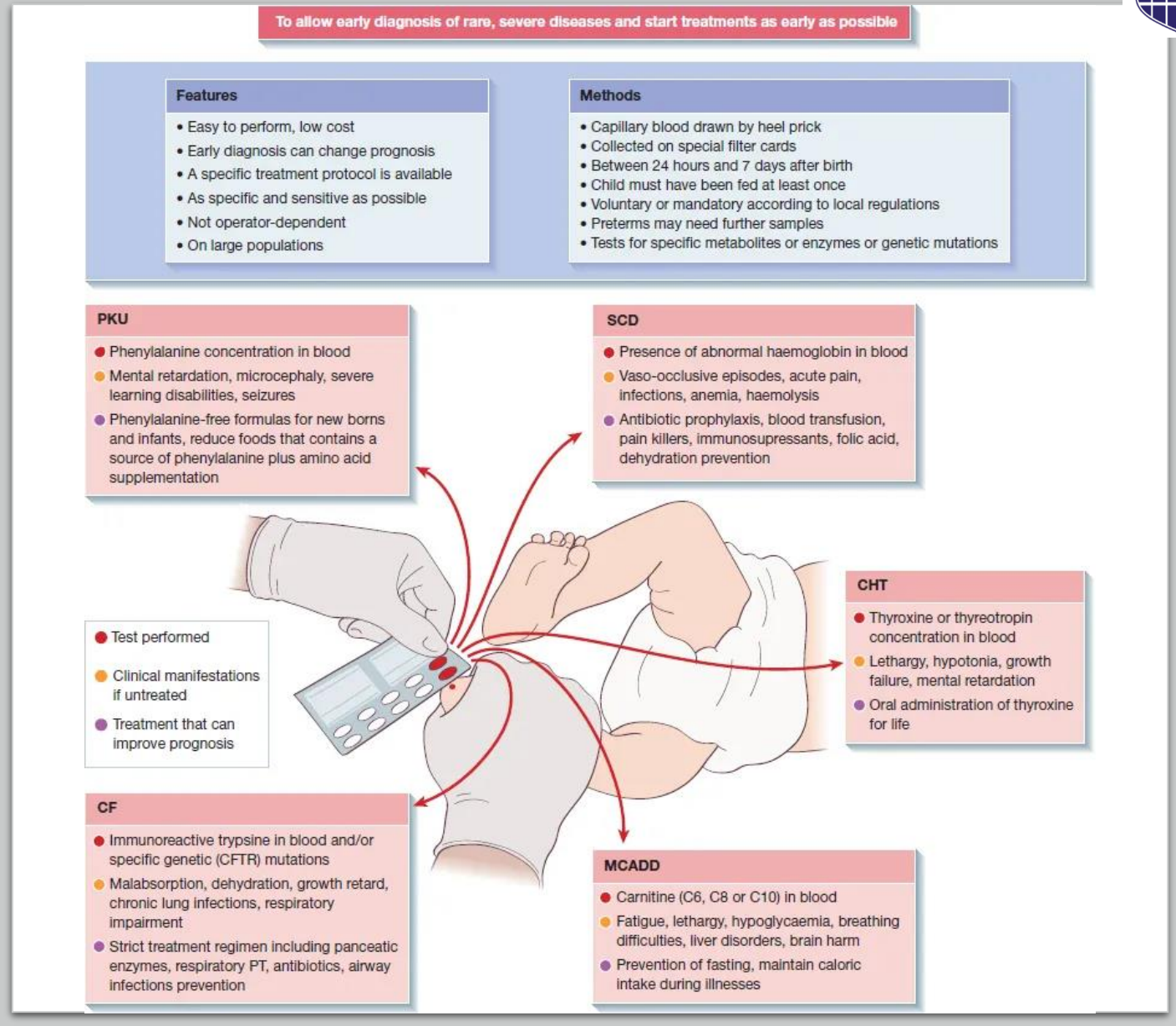
Triple-X Syndrome

- Female.
- Trisomy X; extra X chromosome at pair 23.
- Taller than average.
- No unusual physical features.
- Increased risk of learning disabilities and delayed development of speech and language skills.
- Weak muscle tone.
- Behavioral and emotional difficulties.

Turner syndrome, a condition that affects only females, results when one of the X chromosomes (sex chromosomes) is missing or partially missing. Turner syndrome can cause a variety of medical and developmental problems, including short height, failure of the ovaries to develop and heart defects.

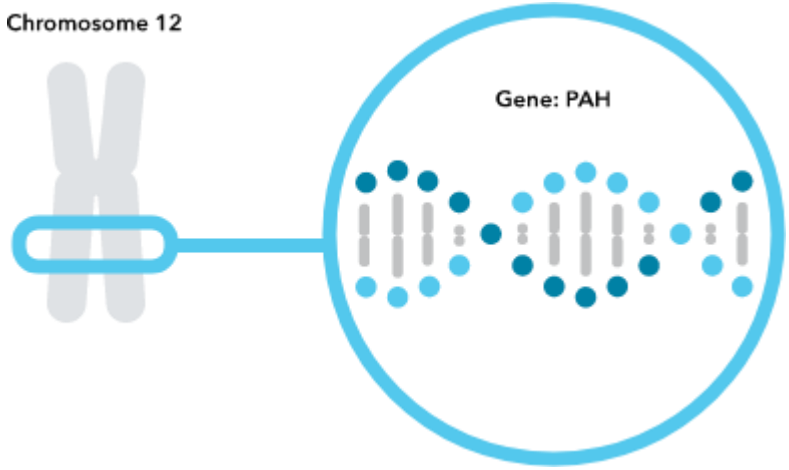


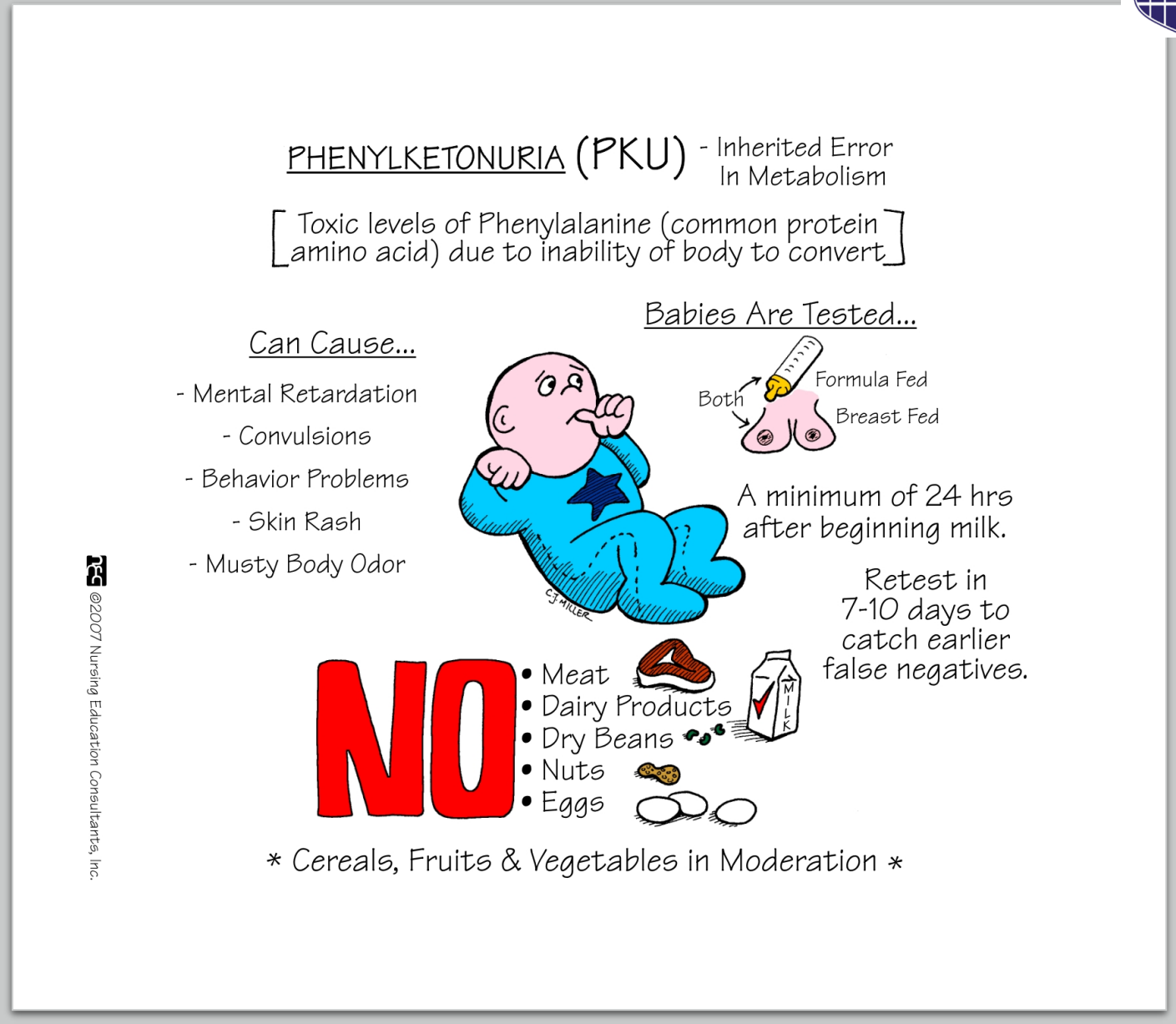
Neonatal screening program (NSPs) are performed on newborn infants to allow the early detection of uncommon and severe congenital diseases that are not clinically evident at birth but in which the prognosis can be significantly influenced by starting specific treatments very early in life of a subject.



Phenylketonuria, also called PKU, is a rare inherited disorder that causes an amino acid called phenylalanine to build up in the body. PKU is caused by a defect in the gene that helps create the enzyme needed to break down phenylalanine.

- This condition is inherited in an autosomal recessive pattern



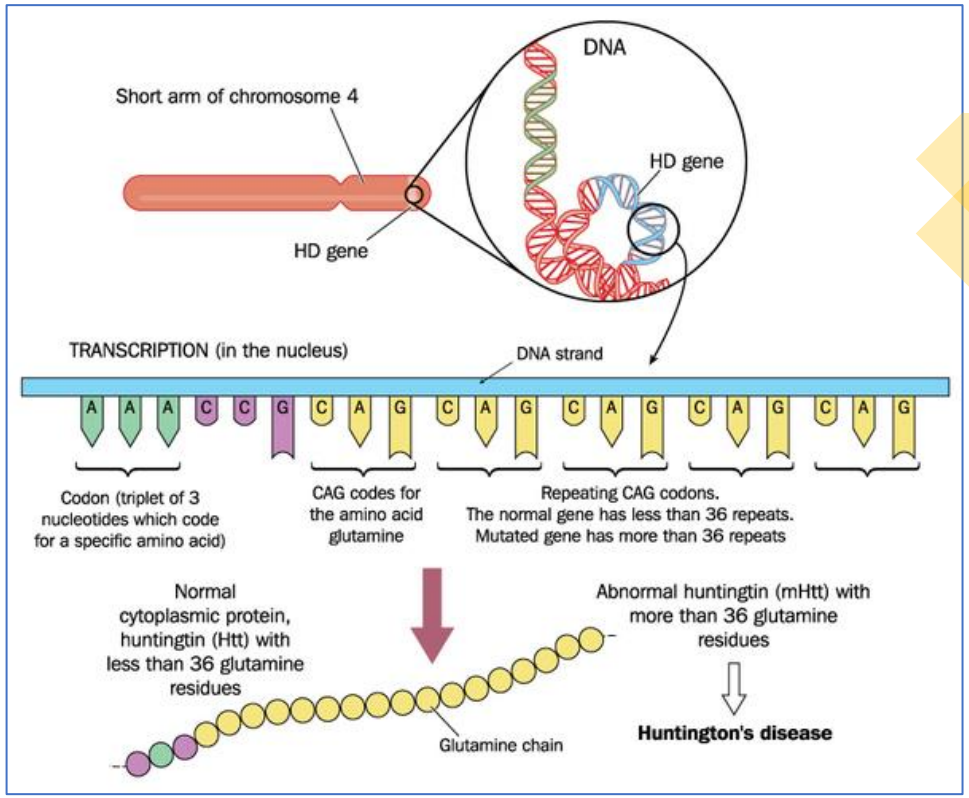


Monogenic Disorders:

- Huntington Disease
- Alzheimer Disease
- Hemochromatosis
- Familial Hypercholesterolemia
- Polycystic Kidney Disease
- Thalassemia

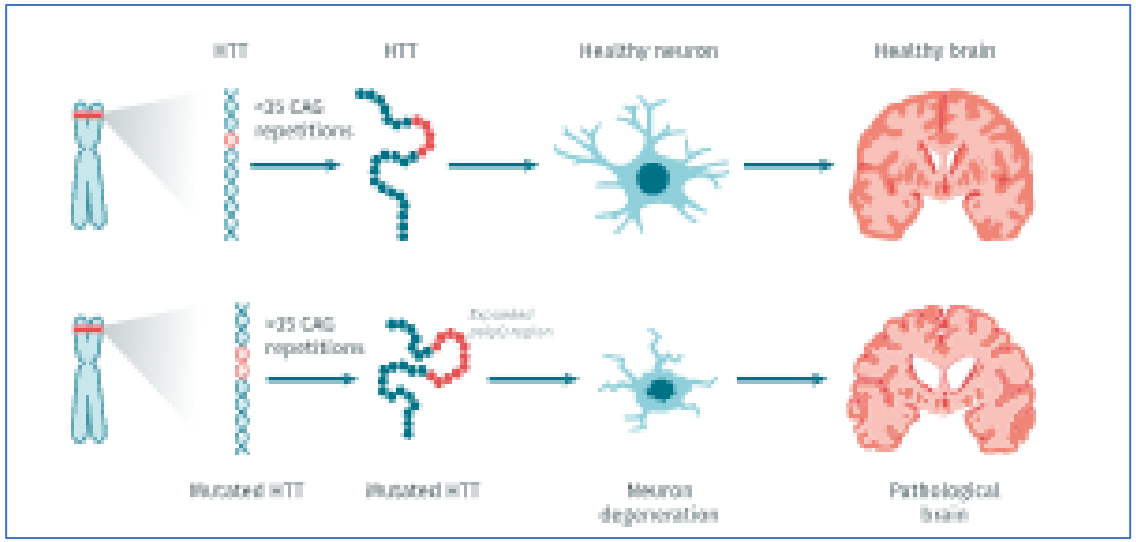
Huntington's disease is an inherited disease characterized by the progressive loss of brain and muscle function.

- Symptoms usually begin during middle age.
- The disease is inherited as an autosomal dominant trait, meaning that a single mutated copy of the responsible gene is sufficient to cause the disease.

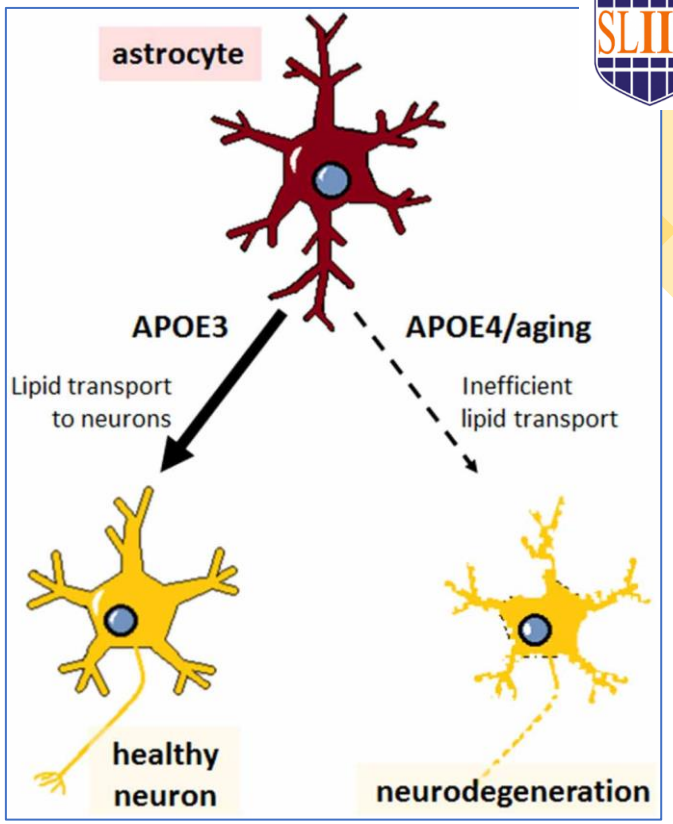


- Mutations in the HTT gene cause Huntington disease. The HTT gene provides instructions for making a protein called huntingtin.

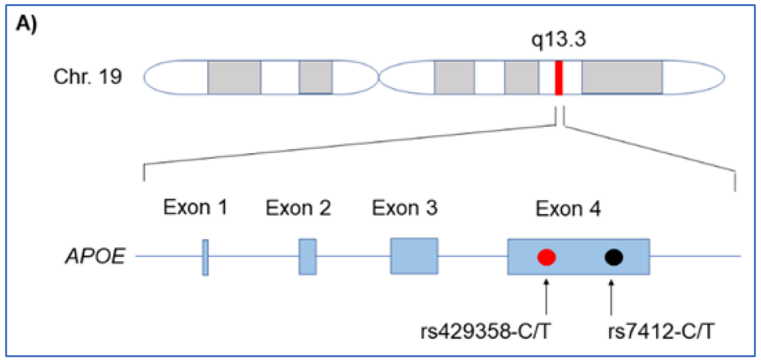
- Although the function of this protein is unclear, it appears to play an important role in nerve cells (neurons) in the brain.

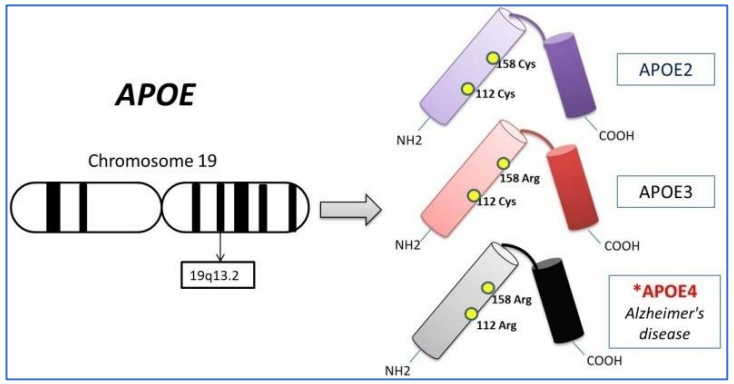


Alzheimer's disease is a progressive neurologic disorder that causes the brain to shrink (atrophy) and brain cells to die.



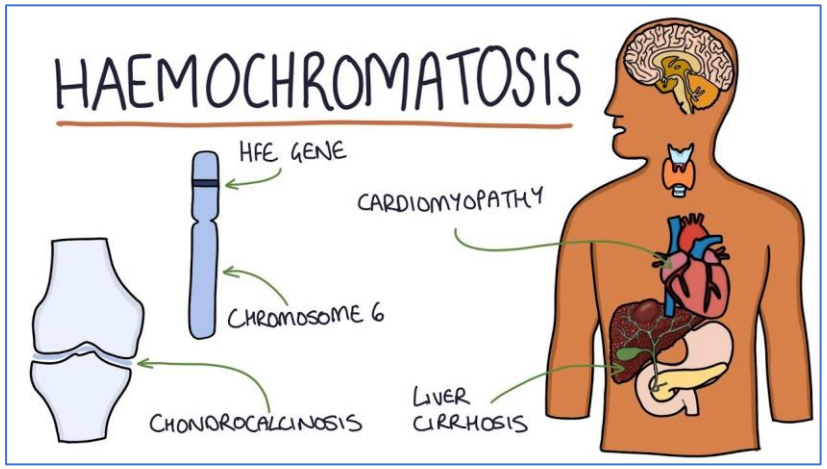
- Alzheimer's disease is the most common cause of dementia — a continuous decline in thinking, behavioral and social skills that affects a person's ability to function independently.
- The most common type of Alzheimer's disease usually begins after age 65 (late-onset Alzheimer's disease).
- The most common gene associated with late-onset Alzheimer's disease is a risk gene called apo-lipo-protein E (APOE).
- APOE has three common forms: APOE e2 — the least common — reduces the risk of Alzheimer's.





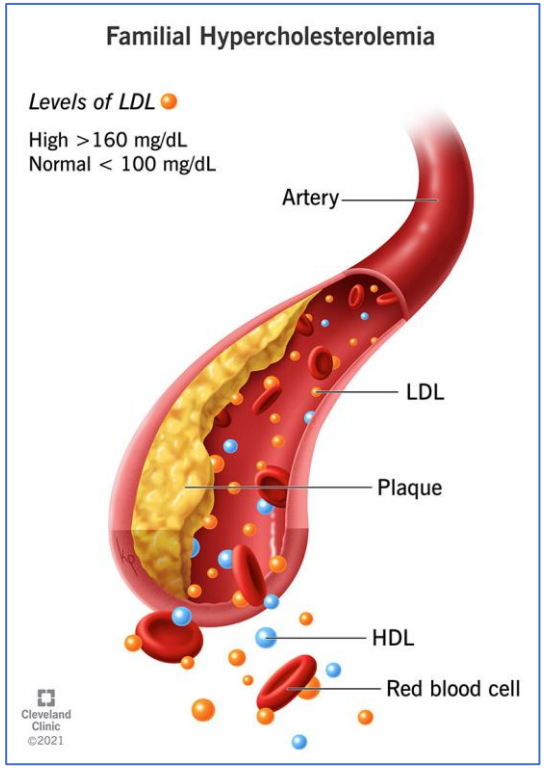
Hemochromatosis is a disorder in which extra iron builds up in the body to harmful levels.

- Without treatment, hemochromatosis can cause iron overload, a buildup of iron that can damage many parts of the body, including your liver, heart, pancreas, endocrine glands, and joints.
- The HFE gene makes a protein involved in iron absorption and causes the body to take in excess iron from food and water.
- In healthy iron metabolism, this gene helps control and regulate how much iron absorbs into our system. There is a feedback loop that is supposed to tell the body to stop taking in iron once it has had enough.
- In hemochromatosis, however, the HFE variations cause a signal that continually and perpetually tells the digestive system to absorb iron from our food. It's like there's no "off switch."



Familial hypercholesterolemia (FH) is a genetic condition which results in a high cholesterol level and subsequently leads to a higher risk of early heart disease.

- It affects approximately 1 in 250 people with around 34 million cases worldwide.
- The condition begins at birth and can cause heart attacks at an early age. Related topics include: Familial combined hyperlipidemia.
- Familial hypercholesterolemia is inherited in families in an autosomal dominant manner. In autosomal dominant inherited conditions, a parent who carries an altered gene that causes the condition has a 1 in 2 (50 percent) chance to pass on that altered gene to each of his or her children.



Familial hypercholesterolemia

- Genetic loci involved in ADH include the LDLR, which codes for the LDL receptor (LDLR), APOB, which codes for apolipoprotein B-100 (apoB-100), the major protein component of LDL, PCSK9, which codes for Proprotein Convertase Subtilisin/Kexin Type 9 (PCSK9), the low abundance circulatory protein that terminates the lifecycle of the LDLR, and apolipoprotein E (APOE), which synthesizes the main protein of triglyceride rich lipoproteins.

Polycystic kidney disease (PKD) is an inherited disorder in which clusters of cysts develop primarily within your kidneys, causing your kidneys to enlarge and lose function over time.

- Cysts are noncancerous round sacs containing fluid. The cysts vary in size, and they can grow very large.

- Autosomal dominant polycystic kidney disease is genetically heterogeneous and associated with mutations in PKD1 (responsible of ADPKD-Type I), PKD2 (-Type II), and GANAB. PKD1 is a complex gene mapping to chromosome 16.

- DNA testing is available for PKD. There are two types of DNA tests: Gene linkage testing and direct mutation analysis/DNA sequencing. Gene linkage can determine if you have PKD with a 99 percent probability in those with family history.

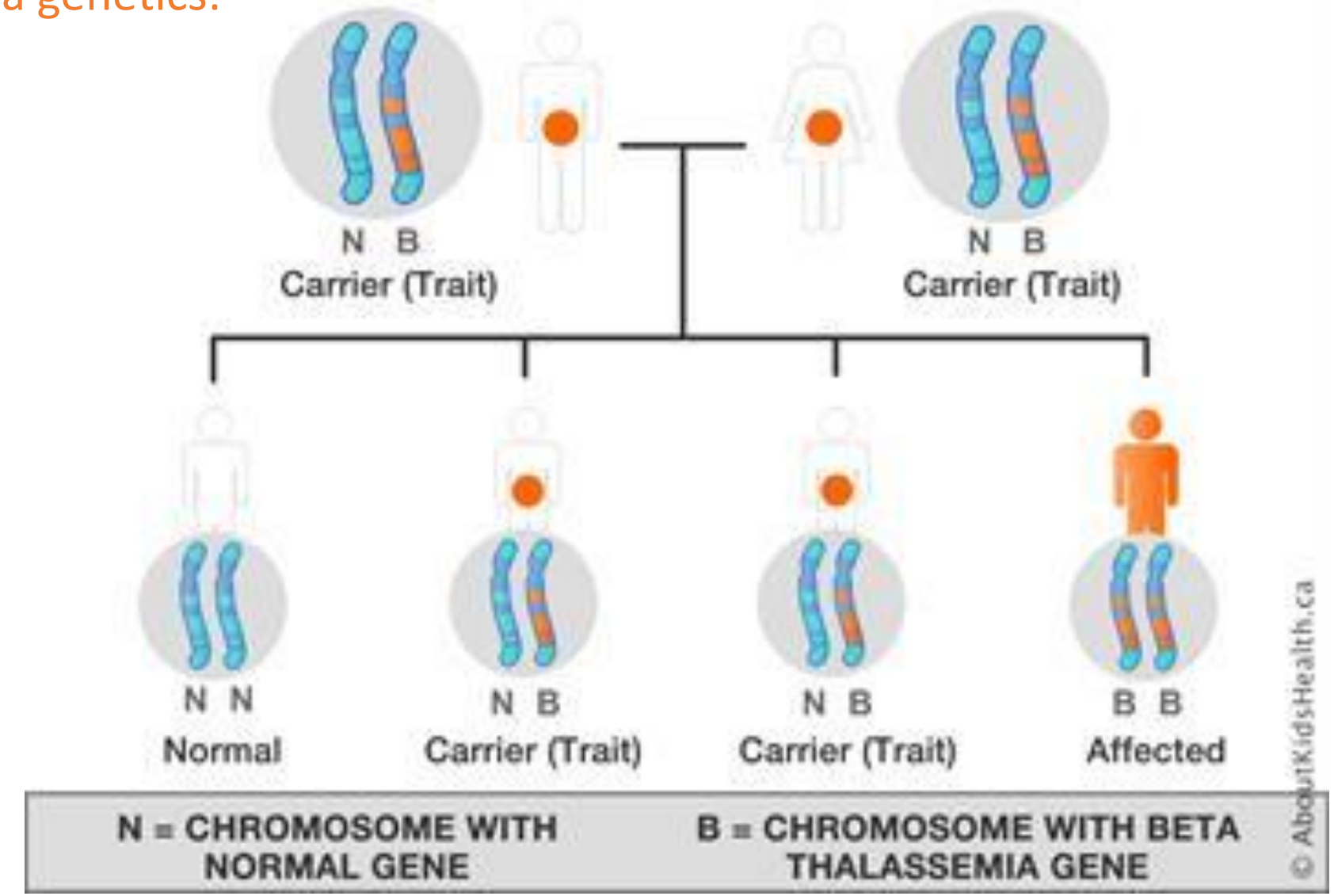
Thalassemia:

- Thalassemia is an inherited blood disorder that reduces the production of functional hemoglobin.
- There are two main types of Thalassemia, alpha Thalassemia and beta Thalassemia.
- Signs and symptoms vary but may include mild to severe anemia, paleness, fatigue, yellow discoloration of skin (jaundice), and bone problems.
- Beta Thalassemia is caused by changes in the HBB gene while alpha Thalassemia is caused by genetic changes in the HBA1 and/or HBA2 genes.
- Both are inherited in an autosomal recessive manner.

- Thalassemia can be diagnosed via a complete blood count, hemoglobin electrophoresis or high-performance liquid chromatography, and DNA testing.

- Bone-marrow transplantation may offer the possibility of a cure in young people who have an HLA-matched donor.

Thalassemia genetics:



Testing for Multifactorial Genetic Disorders

- Coronary Heart Disease
- Hypertension
- Cancers of Complex Origin
- Diabetes
- Rheumatoid Arthritis
- Infectious Diseases
- Psychiatric Diseases

Coronary Heart Disease

- Coronary heart disease is a common cause of morbidity and death. Various genes are often predisposing factors and are better understood than the genetic etiology of many other complex disorders.

- In most cases, no single gene can be identified; a variety of genes (mostly affecting lipid-related proteins) as well as environmental factors such as high-fat and cholesterol diets have been defined (Motulsky and Brunzell, 1992).

- Most affected persons will carry a combination of the underlying predisposing genes, but the specific set of genes accounting for susceptibility is likely to vary from family to family.

Coronary Heart Disease

- However, cholesterol levels alone remain an imperfect predictor of coronary heart disease since many patients with ''normal" cholesterol levels develop coronary heart disease.

- Furthermore, many individuals with high cholesterol levels will not be clinically affected.

- Other predictors of coronary heart disease are already known, such as low levels of high-density lipoprotein (HDL) and high lipoprotein levels.

- High-risk individuals in the population could, therefore, be identified who could take preventive actions by dietary and possibly pharmaceutical interventions.

Hypertension (high blood pressure)

- Multiple genes interacting with environmental factors are likely to be involved, since monogenic inheritance is not observed.
- Hypertension does not exist in individuals in primitive populations but may develop when such individuals are translocated to a Western environment with more salt intake and stress.
- Salt sensitivity (i.e., raised blood pressure associated with salt intake) may be under genetic control.
- Hypertension is a risk factor for strokes, kidney failure, heart failure, and certain eye diseases.

Hypertension (high blood pressure)

- Genetic tests that predict the development of hypertension would be useful once preventive measures to avoid high blood pressure are clearly defined.

- Medications that lower blood pressure are currently used empirically, without defining the specific cause of high blood pressure.

- There is now some evidence that hypertension among African-Americans is less likely to respond to certain classes of drugs (beta-blockers) than hypertension among Caucasians, suggesting genetic heterogeneity in the mechanism of hypertension in these populations. Much more work will be needed before genetic tests can be useful as practical tools for predicting high blood pressure and directing therapy.

Cancers of Complex Origin

- Most cancers have complex etiology; they are caused by alteration of the genetic material (DNA) in somatic cells of various organs (such as breast and colon) that occur after birth.

- Most cancers, therefore, may be classified as somatic genetic disease. Unlike classic heritable diseases where every body cell carries the altered DNA, the characteristic molecular and cellular alterations in cancer affect only the descendants of the original cancer cell in a given organ, which grow in an unregulated fashion and may spread to other organs.

- Detection of the somatic mutations of cancer is becoming increasingly possible by genetic techniques, and many of these techniques are already coming into clinical use.

Cancers of Complex Origin

- The genes affected by these cytogenetic anomalies are being identified and can sometimes be utilized for molecular diagnosis. Occasionally, detection of a specific chromosomal or molecular abnormality is helpful in predicting the clinical severity of the cancer and in a few cases may aid in selecting the most appropriate treatment (IOM, 1992).

- In colon cancer and probably in many other cancers, a sequential series of different mutations of the involved somatic tissue is required for the ultimate development of the malignant tumor (Fearon and Vogelstein, 1990).

- Mutations of common tumor suppressor genes such as the p53 gene are often involved (Frebourg and Friend, 1992). Screening of the general population for common somatic cancers for which everyone is at risk may become feasible using such tumor biomarkers by examining DNA from cells in stool, blood, sputum, and urine, and in breast, cervical, and prostatic fluids.

Diabetes

- Two principal varieties of diabetes exist: early-onset or insulin-dependent diabetes, and late-onset or non-insulin-dependent diabetes (Rotter et al., 1992). Both types have unrelated genetic determinants.

- Insulin-dependent diabetes usually starts early in life and is associated with severe insulin deficiency. Genes of the HLA complex on chromosome 6 have been implicated as etiologic factors in this form of diabetes associated with autoimmune destruction of the pancreas; other genes are likely to be involved but have not been clearly elucidated.

- However, even in family testing, 75 percent of siblings identified to be at risk by HLA testing never develop clinical diabetes. Although preventive therapy of high-risk persons by immunosuppressive drugs is being considered, the drugs currently available are too toxic for general use.

Diabetes

- Non-insulin-dependent diabetes tends to manifest in middle age and is usually milder clinically. This form of diabetes has much stronger genetic determinants, based on identical twin studies in which concordance is almost 100 percent. Unfortunately, the specific genes causing the disease are unknown.

- Since transmission patterns do not fit monogenic inheritance, different interacting genes are likely to be at work; it is likely that different genes are operative in different families, and genes affecting the biochemical action of glucose and insulin are being investigated (Leahy and Boyd, 1993; Mueckler, 1993).

Rheumatoid Arthritis

- Rheumatoid arthritis is a common, often disabling disease that in severe cases can shorten life expectancy markedly (King et al., 1992; Winchester, 1992).
- Certain genes in the HLA system help to determine susceptibility, and other HLA alleles help to determine the severity of the illness.
- However, other poorly defined genetic and environmental factors are also involved in the etiology of rheumatoid arthritis.

Infectious Diseases

- Genetic host factors affecting susceptibility and resistance to many different microbial infections, including viruses, have been recognized for many years in experimental animals (Childs et al., 1992).

- Genetic variation also occurs in the offending microorganism. The effect of genetic variation in both host and microorganism on mortality has become well recognized, although mechanisms governing susceptibility and resistance are often specific for a given infection and may not be related to immunity.

- Twin and adoption studies show some evidence of genetic influences in infectious disease mortality. Genetic factors affecting cellular HLA immunity have been implicated in many different infections. Associations of HLA alleles have been claimed for immunologic responses to a variety of vaccines (e.g., tetanus, influenza, hepatitis A and B) and for infectious diseases (tuberculosis, leprosy, measles, AIDS, and malaria).

Infectious Diseases

- It has already been shown that HLA alleles are involved in influencing the course of HIV (human immunodeficiency virus) infection.

- The interplay of genetically influenced immunologic responses together with more specific genetic variation affecting inborn resistance could affect the natural history of HIV infections.

- The search for genetic variation among appropriate populations (e.g., persons with short versus long latency periods, individuals resistant to HIV infection despite exposure, and identical and fraternal twins) may ultimately uncover the specific mechanisms, and such findings might be useful in devising novel approaches to prevention and therapy in the future.

Psychiatric Diseases

- The most common psychiatric diseases are schizophrenia and manic depressive disorders.

- Based on extensive family, twin, and adoption studies, there is agreement that genetic factors play an important role in the causation of these diseases (Goldin et al., 1992; Hanson and Gottesman, 1992).

- However, the nature and number of the underlying genes are entirely unknown. Mendelian inheritance is rarely observed, and the mechanisms of transmission are generally obscure. There was much excitement when genetic linkage studies using anonymous DNA markers appeared to map specific genes in schizophrenia (Sherrington et al., 1988) and manic depressive disorders (Egeland et al., 1989), but repeated tests to replicate this work have, to date, yielded negative results (Kelsoe et al., 1989; Baron et al., 1993).

Psychiatric Diseases

- Some other neuropsychiatric disorders in which genetic factors have been implicated include panic disorders (Crowe, 1992), Tourette syndrome, and certain types of alcoholism (Propping, 1992). However, no specific genes have been mapped or otherwise identified in any of these disorders.

- It is likely that multiple genes, often interacting with yet poorly understood environmental factors, will be operative in many psychiatric disorders. As with other complex conditions, predictive testing in psychiatric diseases is unlikely to be as accurate as prediction in monogenic diseases. Prediction will always be more probabilistic, and there will be uncertainty regarding whether the disorder will ever manifest and, if so, at what age. The implications of predictive testing for mental disorders raise even more problems than those for other complex medical diseases, because of the heightened potential for stigmatization and discrimination.

Methods for Genetic Testing

- Changes in DNA that lead to altered gene products including single nucleotide variations, nucleotide level insertions or deletions, and generally intragenic variants.

Sanger Sequencing

- Sanger sequencing (chain termination method), first introduced in 1977 works using the selective incorporation of nucleotides that are modified such that they “terminate” the sequencing reaction, thereby identifying the position of the base last incorporated. Sanger sequencing is a mature technology with excellent accuracy and is considered the “gold standard” for sequence determination, although it does not capture mosaicism below 15–20%.

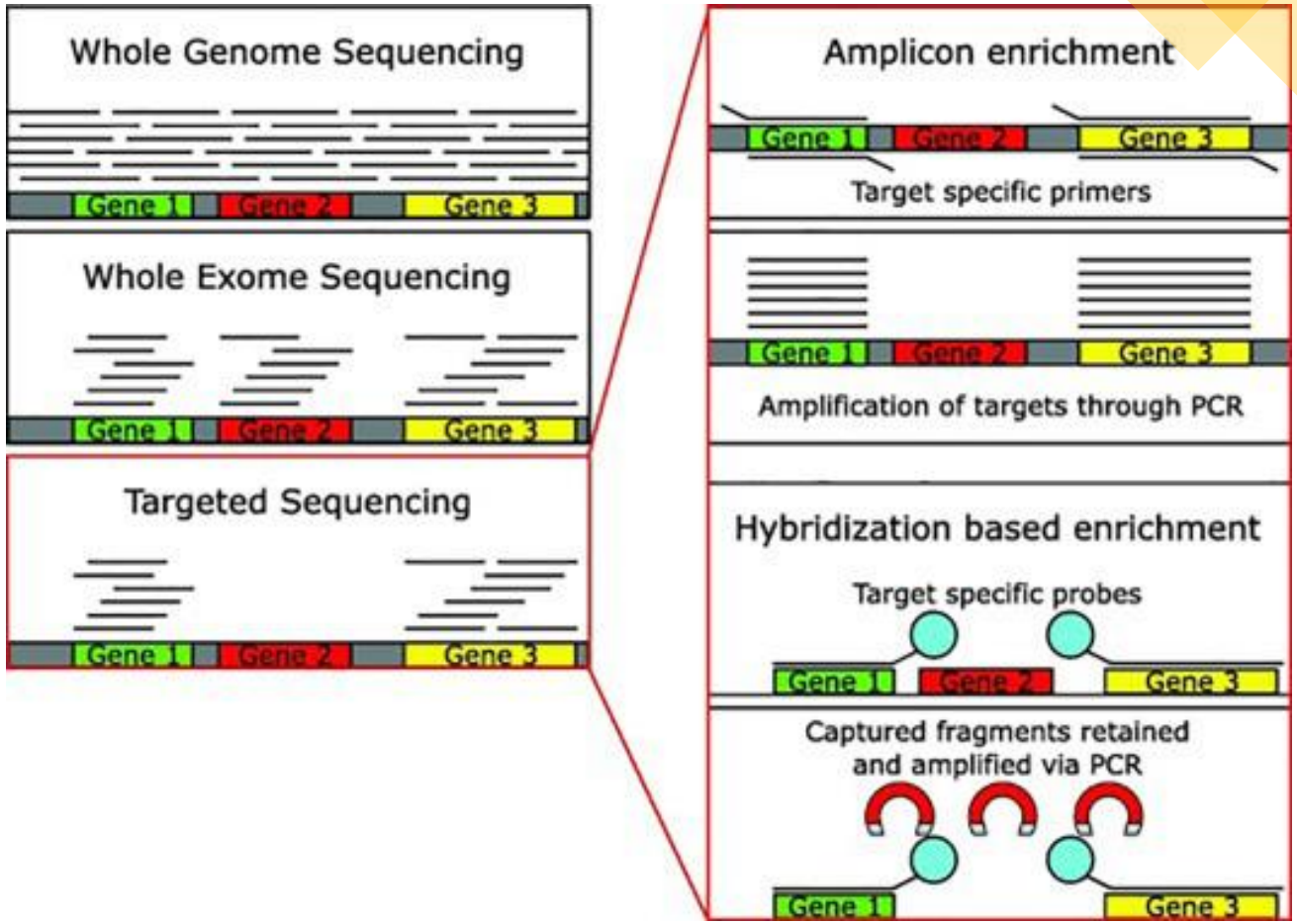
Next Generation Sequencing

- With Illumina sequencing chemistry, for example, sequencing then proceeds by analysis of labeled nucleotides and subsequent imaging, known as “sequencing by synthesis.” Between 2005 and 2012, improvements in sequencing were fast and furious, with increasing accuracy and throughput and decreasing costs, such that beginning around 2011, sequencing of large portions of the genome became possible in the clinical diagnostic laboratory.

    1. Targeted gene panels
    2. Exome sequencing (ES)

Targeted gene panels

- The first clinical application of NGS was to capture and analyze a small groups of genes to look for disease predisposing or disease-causing variants.
- In 2010, this method was applied to analysis of 21 genes involved in breast and ovarian cancer, when Walsh et al., simultaneously analyzed the sequence for these genes following hybridization capture in a cohort of 20 women who had a known mutation in one of the breast and ovarian cancer predisposition genes.



Targeted gene panels:

- This successful proof of principle study laid the foundation for a wide variety of targeted panels that are currently in use in diagnostic laboratories around the world.

- Targeted panels generally analyze between 2 and 500 genes, with different diagnostic strategies depending on details and specificity of the clinical phenotype, the differential diagnosis and the genetic heterogeneity of the phenotype.

Exome sequencing (ES)

- Most disease-causing variants (~85%) are concentrated in the 1–2% of the genome that is protein coding (exonic regions), and the collection of all of the exons is known as the exome.

- These exonic regions can be selectively "captured," allowing sequencing of these high yield genomic regions, reducing the amount of sequence significantly in comparison to the genome, allowing for a relatively efficient test for both new causes of genomic disease or identification of pathogenic variants in known disease genes.
- The diagnostic yield of ES varies between 10 and over 50%, depending on the clinical features of the population tested, year of testing, and analytical strategy.

- For example, diagnostic yield of exome sequencing for chronic kidney disease is 9% based on one recent study, compared to 29–55% for neurodevelopmental disorders, and 26–58% in unselected cohorts with a suspected genetic etiology.

Thank you